# Introduction to Statistics

# Agenda

- ❑ What is Statistics?
- ❑ Data Types & Measurement Level
- ❑ Types of Statistics
  - ➢ Descriptive Vs. Inferential
- ❑ Descriptive Statistics
  - ➢ Types of Descriptive Statistics
  - ➢ Measures of Central Tendency
  - ➢ Measure of Dispersion
- ❑ Population VS. Sampling
- ❑ Inferential Statistics
  - ➢ Data Distribution
  - ➢ Hypothesis Testing
- ❑ Regression
  - ➢ Types of Regression

# What is Statistics?

# What is Statistics?

The science of collecting, organizing, presenting, analyzing, and interpreting data to assist in making more effective decisions.

# Data Types & Measurement Level

# Data Types

Types of data

Categorical

Numerical

Discrete

Continuous

Categorical data represents groups or categories.

**Examples:**

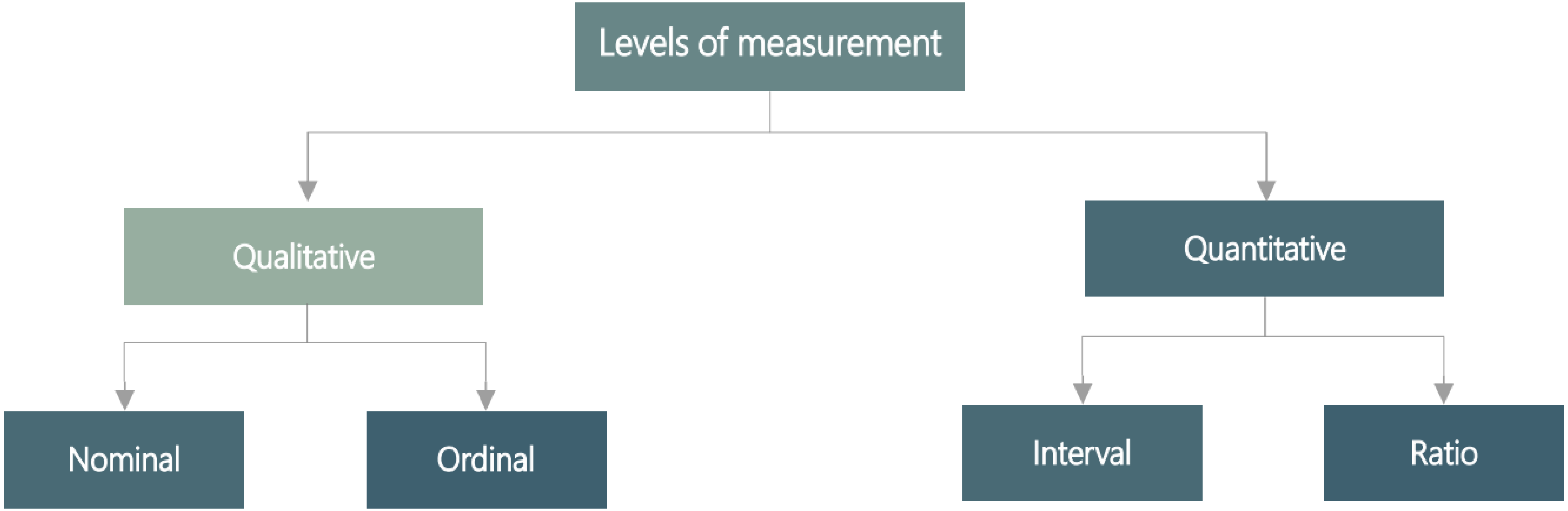
1. Car brands: Audi, BMW and Mercedes.

2. Answers to yes/no questions: yes and no

Numerical data represents numbers. It is divided into two groups: discrete and continuous. Discrete data can be usually counted in a finite matter, while continuous is infinite and impossible to count.

**Examples:**
Discrete: # children you want to have, SAT score
Continuous: weight, height

# Levels of Measurement




There are two qualitative levels: nominal and ordinal. The nominal level represents categories that cannot be put in any order, while ordinal represents categories that **can** be ordered.

**Examples:**
Nominal: four seasons (winter, spring, summer, autumn)
Ordinal: rating your meal (disgusting, unappetizing, neutral, tasty, and delicious)

There are two quantitative levels: interval and ratio. They both represent "numbers", however, ratios **have a true zero,** while intervals don't.

**Examples:**
Interval: degrees Celsius and Fahrenheit
Ratio: degrees Kelvin, length

# Quantitative Variables

**Quantitative variables** are characteristics that can be expressed in **numbers**. For example, weight, height, and length can all be written numerically.

| Name | Weight | Height | Length |
|---|---|---|---|
| Kiara | 4.5 lb | 25 cm | 12 cm |
| Simba | 5 lb | 27 cm | 15 cm |
| Rocky | 4.8 lb | 35 cm | 16 cm |
| Nala | 4.6 lb | 31 cm | 14 cm |
| Bruno | 5 lb | 33 cm | 13 cm |

# Quantitative Variables

- **Continuous Variables**:
  - Contain measurements with decimal precision.
  - **Examples**: Height of individuals, weight of a bag of apples, time taken to run a marathon.
  - **Characteristics**: Measurable, infinitely many possible values, can include fractions/decimals.
- **Discrete Variables**:
  - Contain counts that must be whole integer values.
  - **Examples**: The number of members in a person’s family, or the number of goals a basketball team scored in a game.
  - **Characteristics**: Countable, often integers, clear gaps between values.

# Qualitative Variables

**Qualitative variables** are characteristics of an individual or object which can only be expressed in **words**. Some examples include ethnicity, profession, or gender.

**Ordinal Variables**:

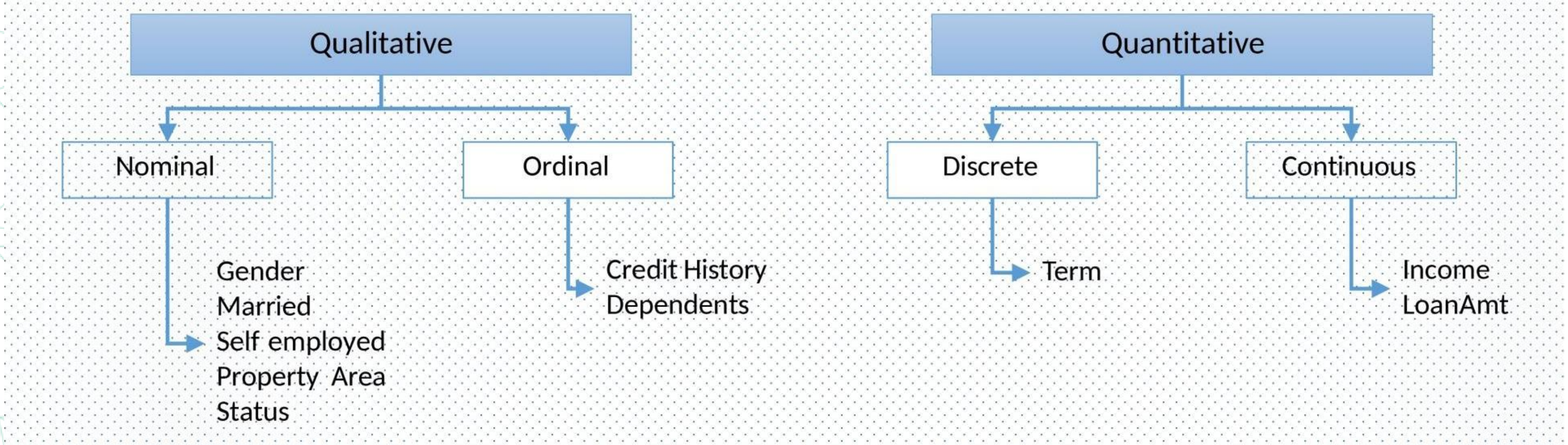
- Variables that are groups containing an inherent ranking.
- **Examples**: Education level (high school, bachelor's, master's), customer satisfaction (dissatisfied, neutral, satisfied).

**Nominal Variables**:

- Variables made up of categories without an inherent order.
- **Examples**: Gender (male, female), eye color (blue, green, brown).

# Understanding the Variables Using Dataset

| Loan_ID | Gender | Married | Dependents | Self_Employed | Income | LoanAmt | Term | CreditHistory | Property_Area | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| LP001002 | Male | No | 0 | No | $5,849.00 | | 60 | 1 | Urban | Y |
| LP001003 | Male | Yes | 1 | No | $4,583.00 | $128.00 | 120 | 1 | Rural | N |
| LP001005 | Male | Yes | 0 | Yes | $3,000.00 | $66.00 | 60 | 1 | Urban | Y |
| LP001006 | Male | Yes | 2 | No | $2,583.00 | $120.00 | 60 | 1 | Urban | Y |



# Understanding the Variables Using Dataset

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| LP001006 | Male | Yes | 2 | No | $2,583.00 | $120.00 | 60 | 1 | Urban | Y |

- **Predictor/Independent**
  - Gender
  - Married
  - Dependents
  - Self_Employed
  - Income
  - LoanAmt
  - Term
  - CreditHistory
  - PropertyArea
- **Target/Dependent**
  - Status

## Data Type

- **Character/String**
  - Gender
  - Married
  - Self_Employed
  - Property_Area
  - Status
- **Numeric**
  - Dependents
  - Income
  - LoanAmt
  - Term
  - CreditHistory

# Types of Statistics

Descriptive Vs. Inferential

# Types of Statistics

- **Descriptive Statistics**
  The methods of organizing, summarizing, and presenting data in an informative way.
  Organizing and summarizing data with frequency tables and frequency distributions.
  Presenting frequency tables and distributions with charts and graphs.
  Measures to summarize the characteristics of data.

- **Inferential Statistics**
  The methods used to estimate population parameters on the basis of a sample statistic. To make inferences (statements) about a population based on a sample, the following concepts are used:
  Probabilities and Probability Distribution
  Sampling
  Estimation
  Hypothesis Testing
  Correlation and Regression

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# Descriptive Statistics

## Types of Descriptive Statistics

# Types of Descriptive Analysis

# Types of Descriptive Statistics

- **Measures of Frequency**:
  - Describe how often certain values or ranges of values occur within a dataset.
  - They are used to understand the distribution and occurrence patterns of the data.

- **Measures of Central Tendency**:
  - These are ways of describing the central position of a frequency distribution for a group of data, such as **mean**, **median**, and **mode**.

- **Measures of Dispersion or Variation**:
  - These are ways of summarizing a group of data by describing how spread out the scores are.
  - Examples include **range**, **interquartile range**, **variance**, **percentile**, **quartile**, and **standard deviation**.

# Frequency Table

- A **frequency table** lists a set of values and how often each one appears.
- **Frequency** is the number of times a specific data value occurs in your dataset.
- These tables help you understand which data values are **common** and which are **rare**.
- They organize your data and are an effective way to present the results to others.
- **Frequency tables** are also known as **frequency distributions** because they allow you to understand the **distribution** of values in your dataset.

| Type of Pet | Frequency |
|---|---|
| Dog | 18 |
| Cat | 12 |
| Fish | 1 |
| None | 10 |

| Heights M | Frequency |
|---|---|
| 1.34 - 1.39 | 4 |
| 1.40 - 1.45 | 14 |
| 1.46 - 1.51 | 31 |
| 1.52 - 1.57 | 19 |
| 1.58 - 1.63 | 14 |
| 1.64 - 1.69 | 6 |

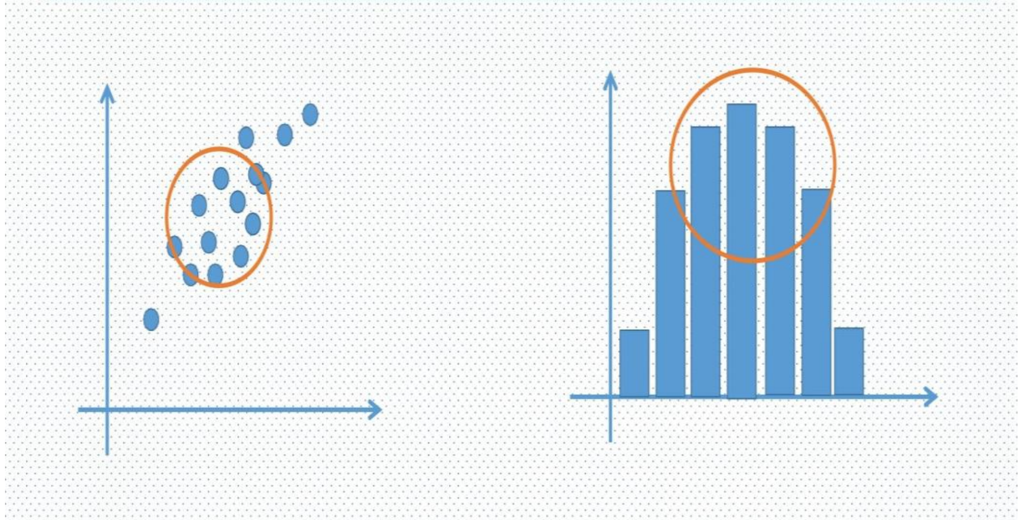
| Satisfaction | Frequency |
|---|---|
| Very satisfied | 59 |
| Satisfied | 42 |
| Neutral | 12 |
| Dissatisfied | 8 |
| Very dissatisfied | 5 |

# Descriptive Statistics

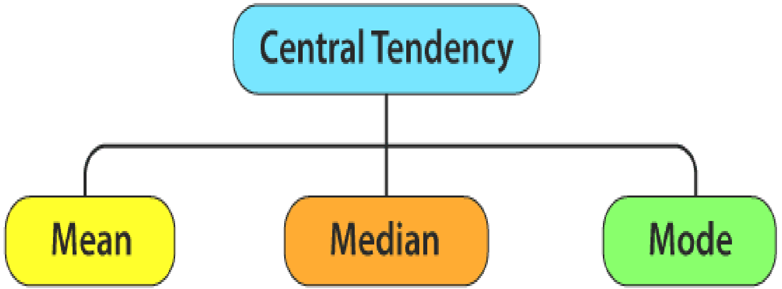
## Measures of Central Tendency

# Central Tendency of Data

- A measure of **central tendency** is a single value that attempts to describe a set of data by identifying the **central position** within that set of data.



- Measures of Central Tendency:



# Mean

- **Mean** is the sum of all the values in the dataset divided by the number of values in the dataset. It is also called the **Arithmetic Average**. **Mean** is denoted as **x̄** and is read as **x bar**.

$$\text{Mean } (\bar{x}) = \frac{\text{Sum of Values}}{\text{Number of Values}}$$

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N}$$

# Mean

The mean is the most widely spread measure of central tendency. It is the simple average of the dataset. **Note: easily affected by outliers**

- **Importance**: The mean provides a central value for a dataset, useful for comparing different data sets and understanding the general trend.

- **Application**: Used in almost all fields such as finance, economics, healthcare, and social sciences to summarize data.

Example Data 1 [ -10, 0, 10, 20, 30 ]   **For data 1:** (-10 + 0 + 10 + 20 + 30 ) / 5 = 10

Example Data 2 [ 8, 9, 10, 11, 12 ]   **For data 2:** (8 + 9 + 10 + 11 + 12 ) / 5 = 10

# Median

- **Median** is the middle value for sorted data. The sorting of the data can be done either in **ascending order** or **descending order**. A **median** divides the data into two equal halves.

- In an ordered dataset, the median is the number at position $\frac{n+1}{2}$ If this position is not a whole number, the median is the simple average of the two numbers at positions closest to the calculated value.

**Example**

Facebook is a popular social networking website. Users can add friends and send them messages, and update their personal profiles to notify friends about themselves and their activities. A sample of 10 adults revealed they spent the following number of hours last month using Facebook.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 3 | 5 | 7 | 5 | 9 | 1 | 3 | 9 | 17 | 10 |

Find the median number of hours.

**Solution**

Note that the number of adults sampled is even (10). The first step, as before, is to order the hours using Facebook from low to high. Then identify the two middle times. The arithmetic mean of the two middle observations gives us the median hours. Arranging the values from low to high:

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 3 | 3 | 5 | 5 | 7 | 9 | 9 | 10 | 17 |

The median is found by averaging the two middle values. The middle values are 5 hours and 7 hours, and the mean of these two values is 6. We conclude that the typical Facebook user spends 6 hours per month at the website. Notice that the median is not one of the values. Also, half of the times are below the median and half are above it.

# Median

The median is the midpoint of the ordered dataset. While It is not as popular as the mean, it is often used in academia and data science because it is not affected by outliers

•**Importance**: The median gives the middle value of a dataset, making it useful for understanding the distribution of data, especially when the data is skewed.

•**Application**: Often used in income data, housing prices, and other scenarios where outliers may skew the mean.

Example :

1, 3, 3, **6**, 7, 8, 9

Median = 6

1, 2, 3, **4**, **5**, 6, 8, 9

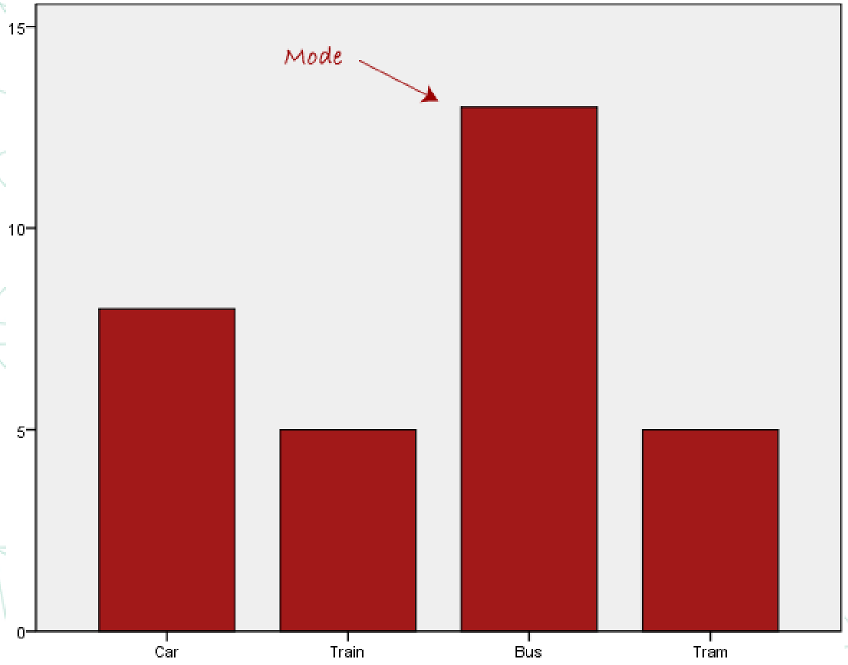
Median = (4 + 5) ÷ 2

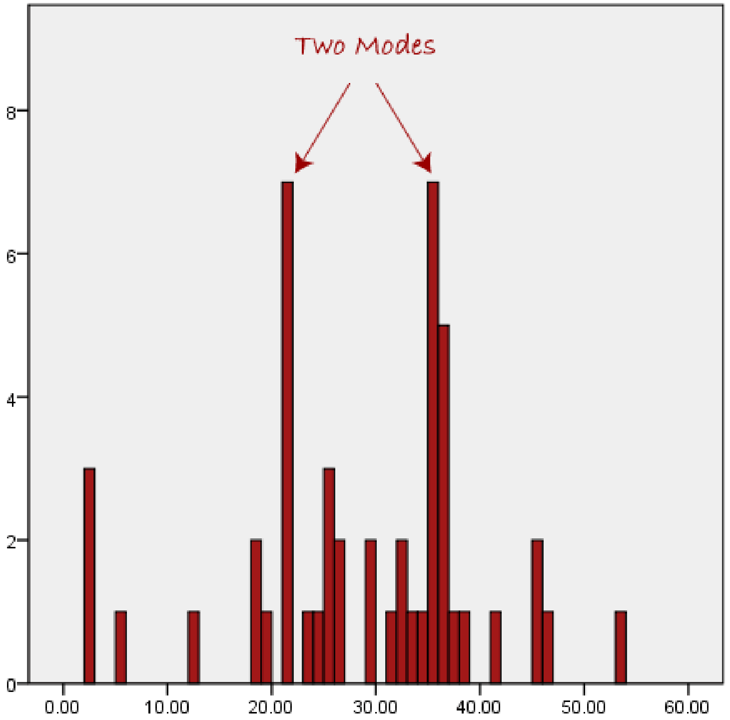
= 4.5

# Mode

- **Mode** is the most frequent value or item in the dataset.
- A dataset can generally have **one** or **more than one** mode value.

Mode = Highest Frequency Term



# Mode

The mode is the value that occurs most often. A dataset can have 0 modes, 1 mode or multiple modes.

The mode is calculated simply by finding the value with the highest frequency.

- **Importance**: The mode indicates the most frequently occurring value in a dataset, useful for understanding the most common occurrences.

- **Application**: Common in market research, quality control, and inventory management.

Example Data 1 [ 10, 13, 15, 16, 13, 15, 11, 13]

Example Data 2 [ 8, 9, 13, 11, 12, 16, 8, 10 ]

**For data 1:** 13 is the most frequent

**For data 2:** 8 is the most frequent

| Applicant | Loan Amount |
|---|---|
| Jitesh | $ 24,000 |
| John | $ 18,000 |
| Frans | $ 34, 000 |
| Danny | $ 40,000 |
| Cecile | $ 24,000 |
| Scott | $ 16,000 |
| Alex | $ 29,000 |

Mode = $24,000

# Differences between Mean, Median and Mode

| Feature | Mean | Median | Mode |
|---|---|---|---|
| Definition | Mean is the average of all values. | Median is the middle value when data is sorted. | Mode is the most frequently occurring value in the dataset. |
| Sensitivity | Mean is sensitive to outliers. | Median is not sensitive to outliers . | Mode is not sensitive to outliers. |
| Calculation | Calculated by adding up all values of a dataset and dividing them by the total number of values in dataset. | Calculated by finding the middle value in a list of data. | Calculated by finding which value occurs more number of times in a dataset. |
| Representation | Value of mean may or may not be in dataset. | Value of median is always a value from the dataset. | Value of mode is also always a value from the dataset. |

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# Effect of Outliers in Descriptive Statistics

- An outlier is any unusually **large** or **small** observation.
- Outliers can have a disproportionate effect on statistical results, such as the mean but doesn't affect median or mode which can result in **misleading impressions**

| Experience | Salary |
|---|---|
| 1 | $ 3,725 |
| 2 | $ 4,155 |
| 3 | $ 4,627 |
| 4 | $ 5,147 |
| 5 | $ 5,718 |
| 6 | $ 6,347 |
| 7 | $ 7,039 |
| 8 | $ 7,210 |
| 9 | $ 7,423 |
| 10 | $ 7,556 |
| 11 | $ 8,369 |
| 12 | $ 8,810 |
| 13 | $ 8,940 |
| 14 | $ 9,200 |
| 15 | $ 9,458 |

6,915 $ ← Mean → 7,676 $

7,200 $ ← Median → 7,200 $

| Experience | Salary |
|---|---|
| 1 | $ 3,725 |
| 2 | $ 4,155 |
| 3 | $ 4,627 |
| 4 | $ 5,147 |
| 5 | $ 5,718 |
| 6 | $ 6,347 |
| 7 | $ 7,039 |
| 8 | $ 7,210 |
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# Effect of Outliers in Descriptive Statistics


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# Descriptive Statistics

## Measure of Dispersion

# Difference between Central Tendency and Dispersion

- **Central tendency** tells you where most of your data points lie, while **dispersion** summarizes how far apart your points are from each other.
- Datasets can have the same **central tendency** but different levels of **dispersion**, or vice versa. Together, they give you a complete picture of your data.


Central Tendency


Spread in data

- This example show how central tendency alone is not enough to describe data here we have same mean and median but each time with different data, so we need to measure dispersion

| Day | Temperature |
|---|---|
| 1 | 20 |
| 2 | 21 |
| 3 | 19 |
| 4 | 20 |
| 5 | 21 |
| 6 | 19 |
| 7 | 20 |
| **Total** | **140** |

Mean : 20

Median : 20

| Day | Temperature |
|---|---|
| 1 | 22 |
| 2 | 23 |
| 3 | 21 |
| 4 | 18 |
| 5 | 19 |
| 6 | 17 |
| 7 | 20 |
| **Total** | **140** |

Mean : 20

Median : 20

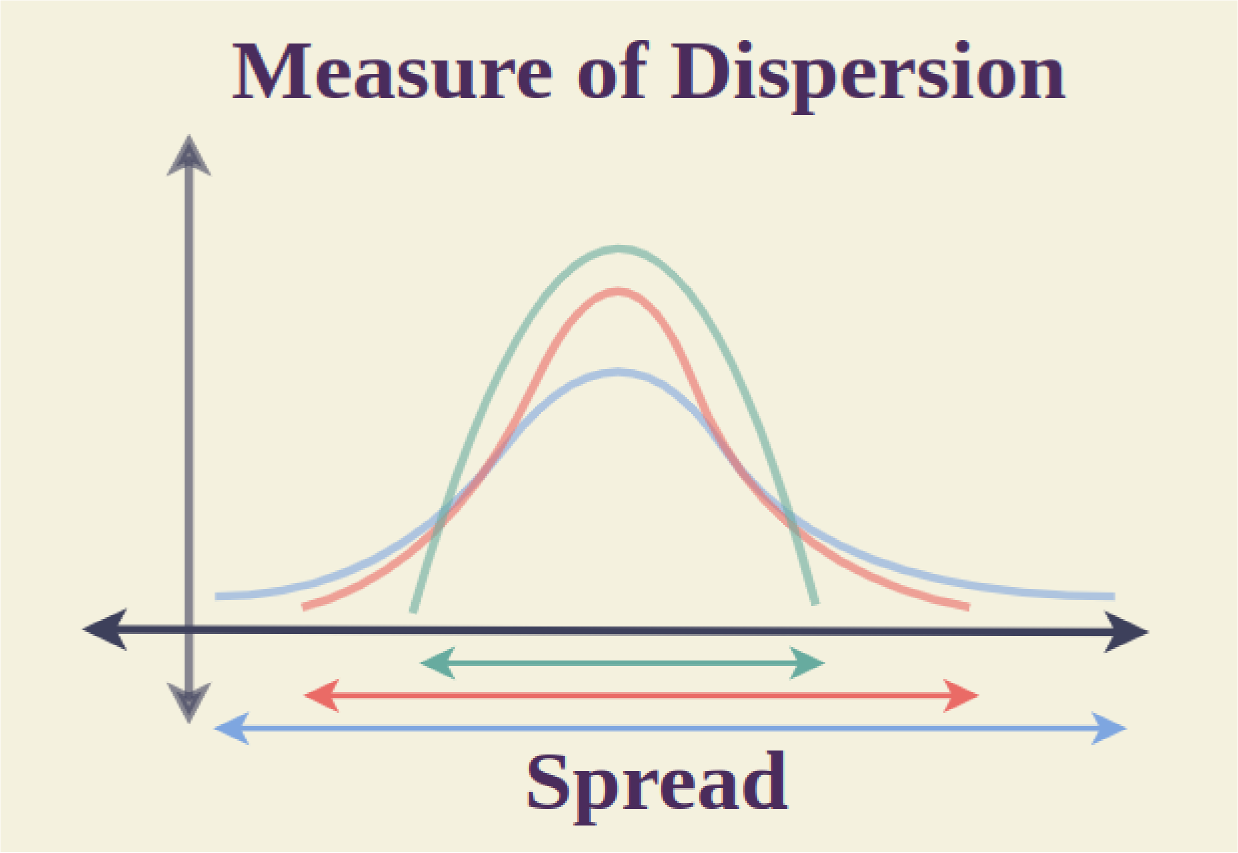
| Day | Temperature |
|---|---|
| 1 | 12 |
| 2 | 11 |
| 3 | 13 |
| 4 | 20 |
| 5 | 24 |
| 6 | 29 |
| 7 | 31 |
| **Total** | **140** |

Mean : 20

Median : 20

# Measure of Dispersion

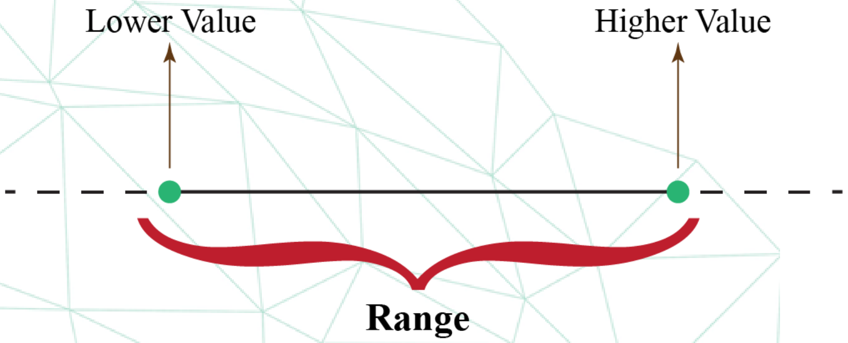
- Range
- Interquartile range
- Percentile
- Variance
- Standard deviation



# Range

- **The range** is the difference between the highest value and the lowest value of the **data**. It helps in knowing the **spread** of the data.

- **Application**: Widely used in everyday scenarios such as evaluating temperature variations over a week, comparing prices of products, and understanding the performance spread of students in a test.




16, 24, 22, 25, 26, 27, 28, 23

**Range = max - min**

**Range = 28 - 16 = 12**

# Question

Calculate the range of the given set of data:
7, 47, 8, 42, 47, 95, 42, 96, 2.

A) 90

B) 94

C) 96

D) 100

# Range and Interquartile Range

- **It** is a better measure of dispersion than **range** because it leaves out the extreme values.

- It equally divides the distribution into four equal parts called **quartiles**:

•The first 25% is the **1st quartile (Q1)**.
•The middle one is the **2nd quartile (Q2)**.
•The last one is the **3rd quartile (Q3)**.

- The **2nd quartile (Q2)** divides the distribution into two equal parts of 50%, so it is the same as the **Median**.

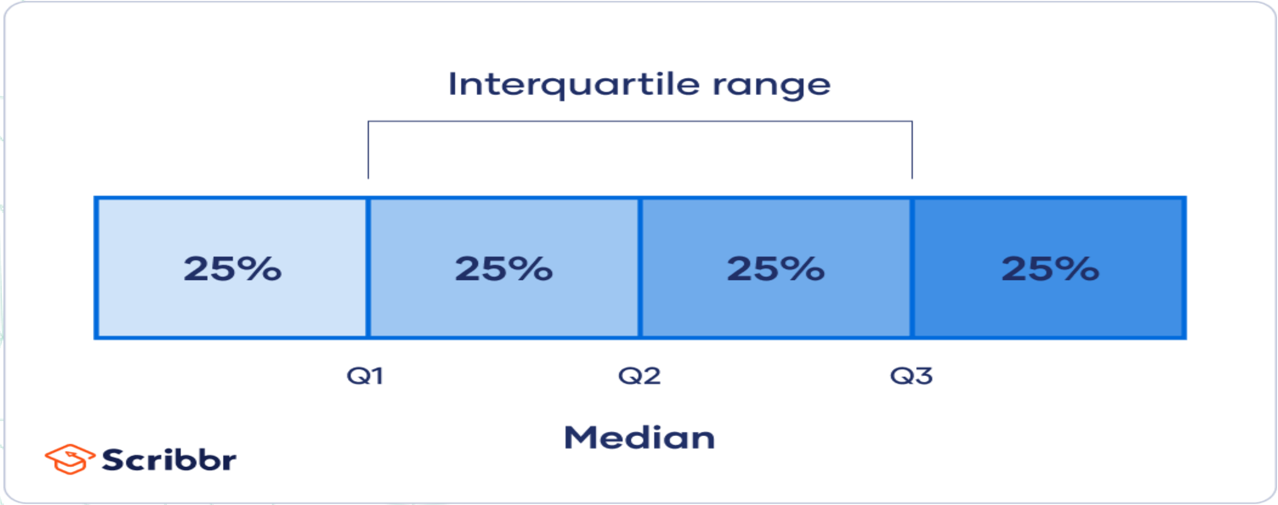
- The **interquartile range** is the distance between the third and the first quartile, or, in other words, **IQR = Q3 - Q1**.

# Range and Interquartile Range

- **The interquartile range** is a measure of where the “middle fifty” is in a data set.

- Where a **range** is a measure of where the beginning and end are in a set, the **interquartile range** is a measure of where the bulk of the values lie.
- That’s why it’s preferred over many other measures of spread when reporting things like school performance or SAT scores.

- The **interquartile range formula** is the first quartile subtracted from the third quartile:
  - **IQR = Q3 – Q1**.



# Quartile Calculations

## Interquartile range (IQR) method

**Calculate IQR:**

To calculate the IQR, you can follow these steps:

- **Sort** the data in **ascending** order.
- Find the **median** of the dataset.
- Divide the dataset into **two halves**, the lower half, and the upper half. If the median is included in either half, include it in **both halves**.
- Find the **median** of the **lower** half (**Q1**) and the **median** of the **upper** half (**Q3**).
- Calculate the **IQR** by subtracting **Q1** from **Q3**: **IQR = Q3 - Q1**.

Example:

9, 3, 2, 5, 7, 8, 6, 1, 4, 10

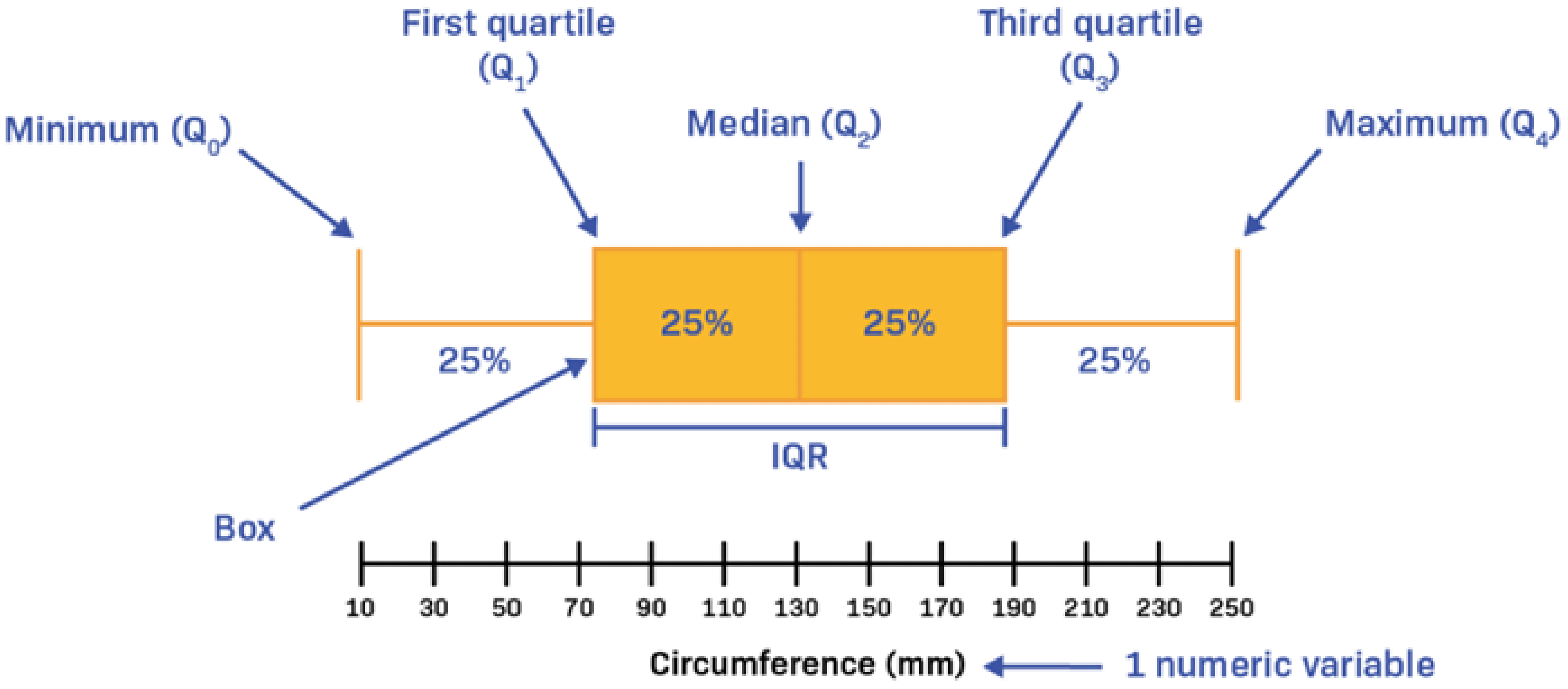
1, 2, 3, 4, **5, 6**, 7, 8, 9, 10 -→ **Q2** or **median** = (5+6)/2 = 5.5

Lower half: 1, 2, **3**, 4, 5 -→ **Q1** = 3

Upper half: 6, 7, **8,** 9, 10 -→ **Q3** = 8

**IQR** = Q3 - Q1 = 8 - 3 = **5**

# Box Plot


First quartile
(Q1)
Third quartile
(Q3)
Minimum (Q0)
Median (Q2)
Maximum (Q4)
25%
25%
25%
25%
IQR
Box
10 30 50 70 90 110 130 150 170 190 210 230 250
Circumference (mm)
1 numeric variable

# Question

Calculate the interquartile range (IQR) of the following data: 17, 18, 18, 19, 20, 21, 21, 23, 25.

A) 4

B) 5

C) 6

D) 7

# Question

Find the interquartile range (IQR) using the following values:

• Minimum: 1
• Q1: 3
• Median: 5
• Q3: 7
• Maximum: 9

A) 2

B) 3

C) 4

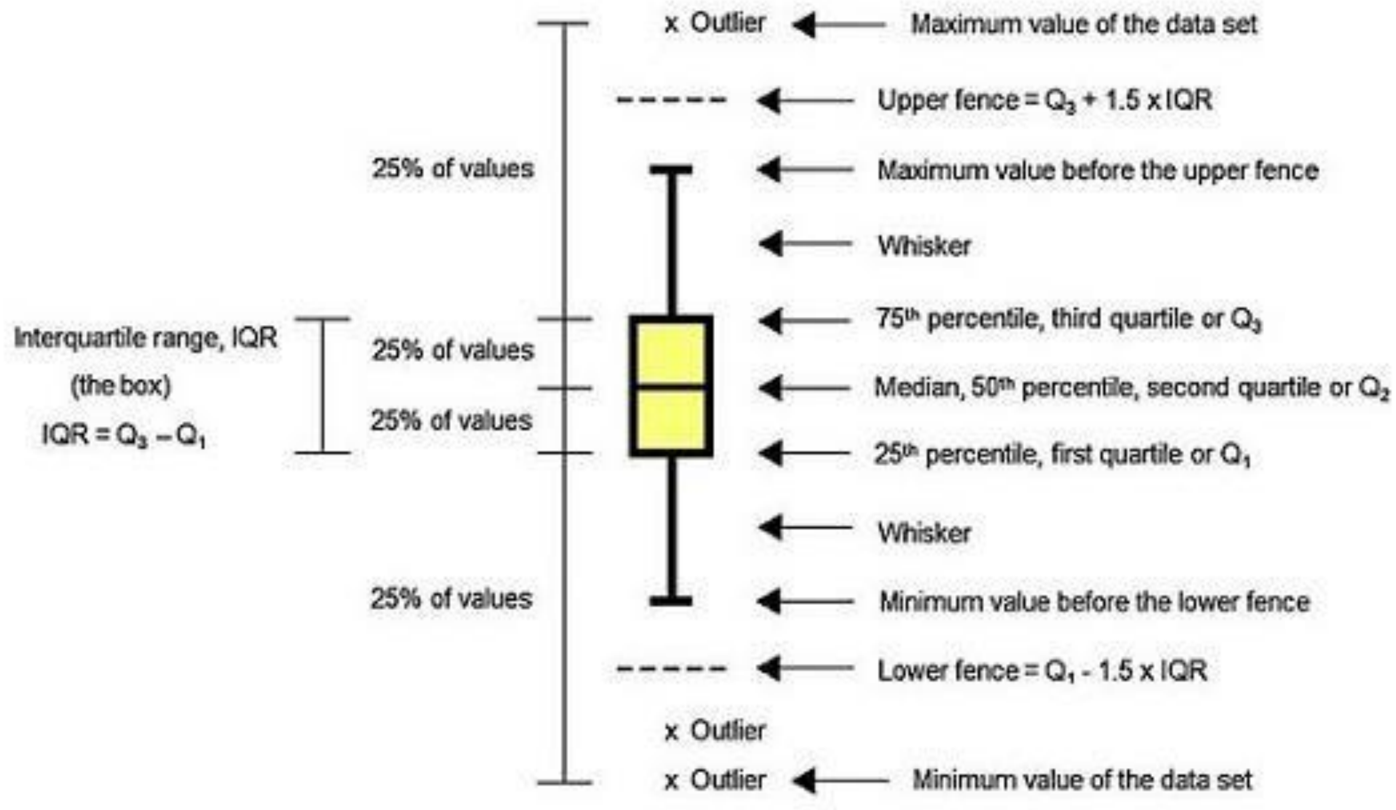
D) 5

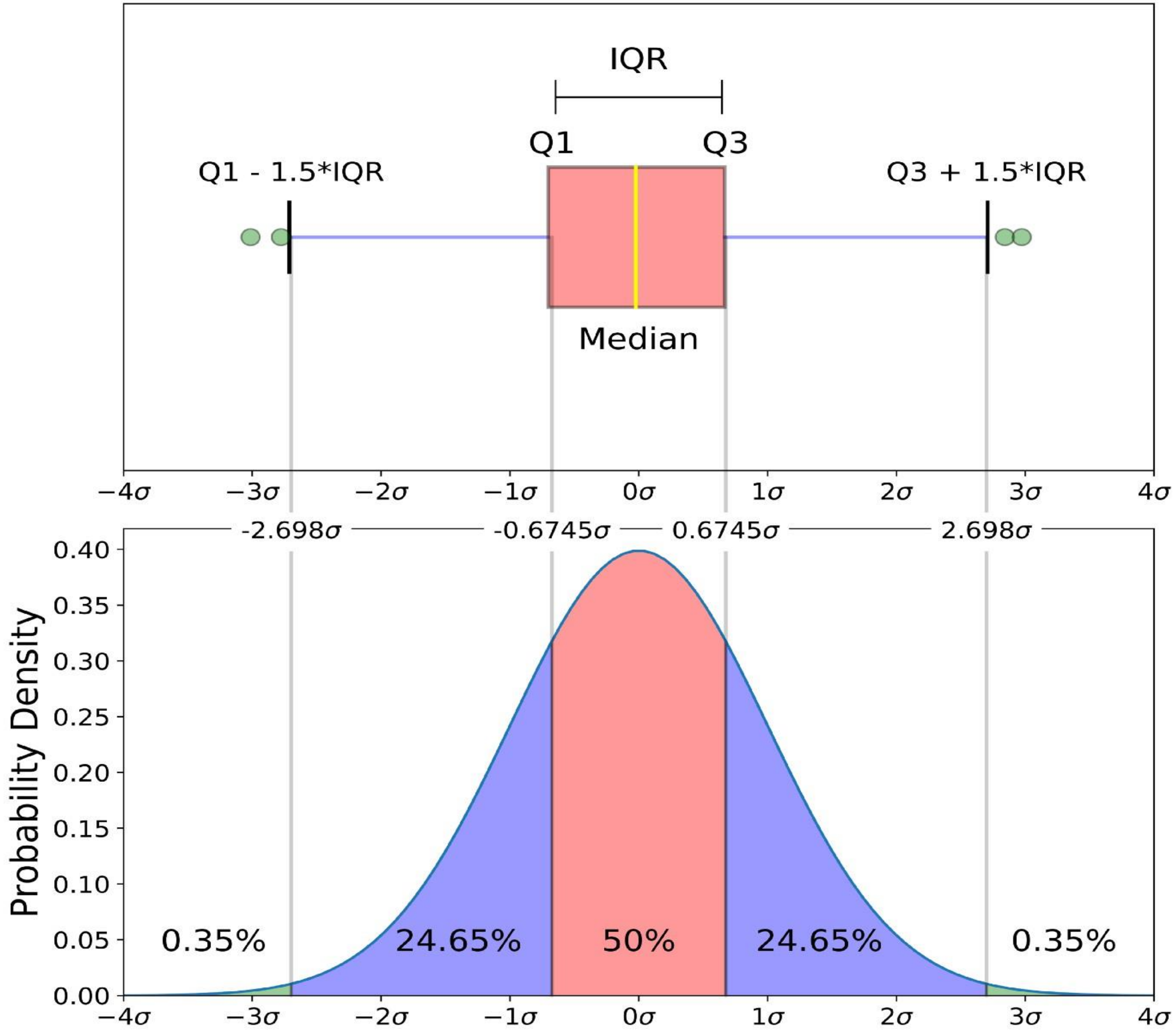
# Advantage of IQR

- The main advantage of the **IQR** is that it is not affected by **outliers** because it doesn't consider observations below **Q1** or above **Q3**.
- Observations can be considered outliers when they lie more than **1.5 IQR** below the first quartile or **1.5 IQR** above the third quartile.
- **Outliers** = **Q1 – 1.5 × IQR** (or) **Q3 + 1.5 × IQR**

# Box Plot

- mainly used when you are describing center and variability of your data.
- It is also useful for detecting outliers in the data.
- Used to visualize IQR






**IQR is plotted in a boxplot and probability density**

# Percentile

- A percentile is a measure used in statistics to indicate the value below which a given percentage of observations in a group of observations fall.

$$Percentile(x) = (\text{Number of values fall under 'x'} / \text{total number of values}) \times 100$$

$$P = (n/N) \times 100$$

*Where ,*

- *P is percentile*
- *n – Number of values below 'x'*
- *N – Total count of population*

# Percentile

- **Arrange** the data in an order
- **Calculate** the **percentage** of observations or data points below a particular value.

What is the **80$^{th}$** Percentile observation?

| Row Number | Salary |
|---|---|
| 1 | $ 3,725 |
| 2 | $ 4,155 |
| 3 | $ 4,627 |
| 4 | $ 5,147 |
| 5 | $ 5,718 |
| 6 | $ 6,347 |
| 7 | $ 7,039 |
| 8 | $ 7,210 |
| 9 | $ 7,423 |
| 10 | $ 7,556 |
| 11 | $ 8,369 |
| 12 | $ 8,810 |
| 13 | $ 8,940 |
| 14 | $ 9,200 |
| 15 | $ 9,458 |

Total Observations * 0.8

15*0.8= 12

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# Percentile calculation

- The number that expresses the value that a given percent of the values are lower than
- Example:
- We have an array of the ages of all the people that working in same office ages = [25,31,43,48,50,41,39,60,52,32,27,46,47,55]
- What is 75. percentile? The answer is 48, meaning that 75% of the people are 48 or younger.
- Steps:
  - Arrange in ascending order [25,27,31,32,39,41,43,46,47,48,50,52,55,60]
  - Find the Rank = Percentile/100 * (number of things )
  - Rank= 0.75*14=10.5
  - take the round 11 then subtract 1 for zero index so rank=10 with value 48

# Variance

- **Variance** is a measure of how far a set of data are dispersed from their **mean** or **average** value. It is denoted as **'$\boldsymbol{\sigma^2}$'**.

**Properties of Variance**:

- It is always non-negative since each term in the variance sum is squared, and therefore, the result is either positive or zero.
- Variance always has **squared units**. For example, the variance of a set of weights estimated in kilograms will be given in **$kg^2$**.
- Since the population variance is squared, we cannot compare it directly with the **mean** or the data themselves.

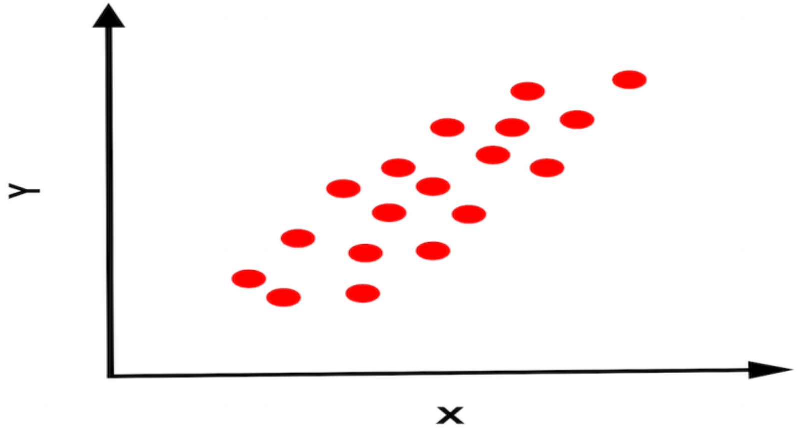
# Variance ($\sigma^2$)

Variance measures the dispersion of a set of data points around its mean value.

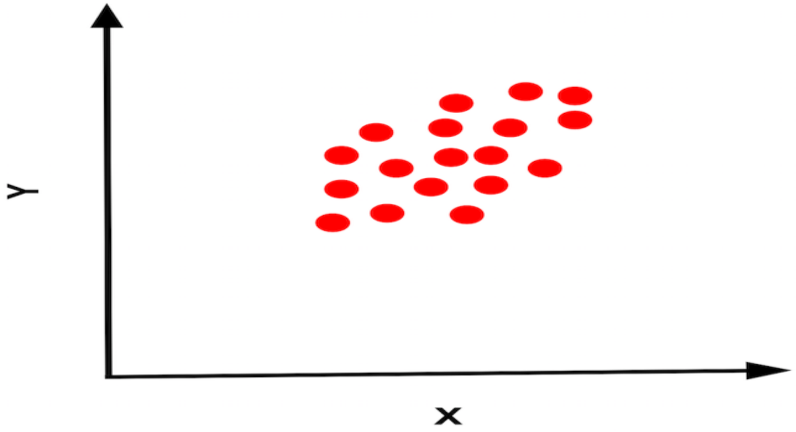
- **Importance**: Variance measures the spread of data points around the mean, useful for understanding data variability.

- **Application**: Used in risk management, investment analysis, and quality control.

In Figure 1, the points have a high variance because they are spread out,




In Figure 2, the points have a low variance because they are close together.



# How to Calculate Variance Step by Step

- **Calculate the Mean** (**$\bar{x}$**): Find the average of all data points.
- **Subtract the Mean from Each Observation** (**$X - \bar{x}$**): For each data point, subtract the mean from the data point.
- **Square Each of the Resulting Observations** (**$(X - \bar{x})^2$**): Square the result of each subtraction.
- **Add These Squared Results Together**: Sum all the squared values.
- **Divide This Total by the Number of Observations** (**n**) (in the case of a population) to Get **Variance** (**$\sigma^2$**): Divide the sum of squared values by the number of observations to obtain the variance.

$$\sigma^2 = \frac{\sum (xi - \bar{x})^2}{N}$$

# Variance and Standard Deviation

| Day | X | $X-\bar{X}$ | $(X-\bar{X})^2$ |
|---|---|---|---|
| 1 | 20 | 0 | 0 |
| 2 | 21 | 1 | 1 |
| 3 | 19 | -1 | 1 |
| 4 | 20 | 0 | 0 |
| 5 | 21 | 1 | 1 |
| 6 | 19 | -1 | 1 |
| 7 | 20 | 0 | 0 |

- Average = 4/7 = 0.57
- σ2 = Variance = 0.57

# Standard Deviation

- **Standard deviation** measures the deviation of data from its mean or average position. The degree of dispersion is computed by estimating the deviation of data points. It is denoted by the symbol **‘σ’**.

- **Properties of Standard Deviation**:

- It describes the square root of the mean of the squares of all values in a data set and is also called the **root-mean-square deviation**.

- The smallest value of the standard deviation is **0** since it cannot be negative.

- When the data values of a group are similar, the standard deviation will be very low or close to zero. However, when the data values vary significantly from each other, the standard deviation will be high or farther from zero.

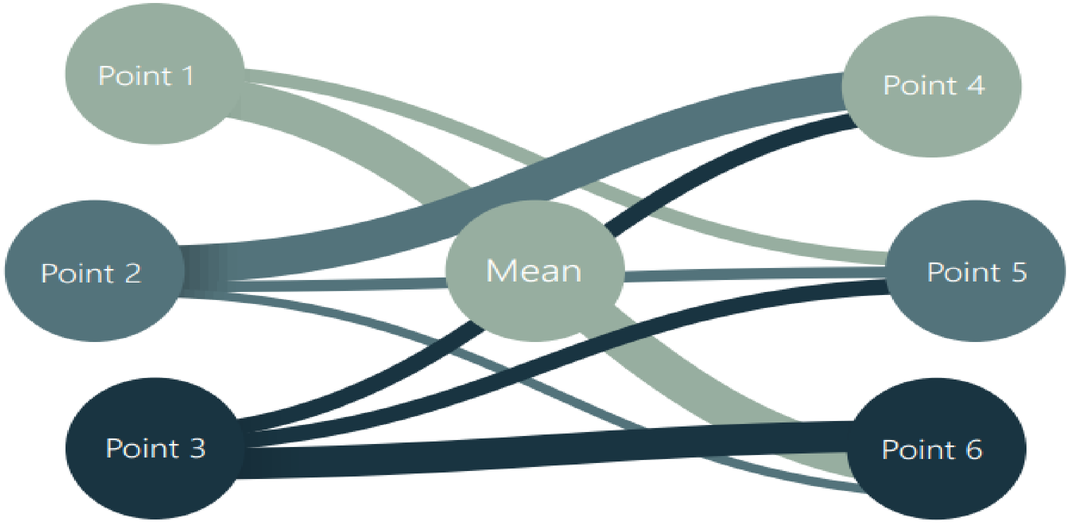
$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

# Standard deviation std (σ)

The std is the most common way to measure the spread of the data and how close data points are to each other.

- **Importance**: Standard deviation is the square root of variance, providing a measure of data dispersion that is in the same unit as the data. $\sigma = \sqrt{\sigma^2}$

- **Application**: Commonly used in finance for assessing investment risk, and in process control for monitoring production quality.



# Variance and standard deviation

| Day | Temperature |
|---|---|
| 1 | 20 |
| 2 | 21 |
| 3 | 19 |
| 4 | 20 |
| 5 | 21 |
| 6 | 19 |
| 7 | 20 |

$\sigma = 0.7559$

$\text{Mean} = \overline{X} = 20$

| Day | Temperature |
|---|---|
| 1 | 12 |
| 2 | 11 |
| 3 | 13 |
| 4 | 20 |
| 5 | 24 |
| 6 | 29 |
| 7 | 31 |

$\sigma = 7.67$

$\text{Mean} = \overline{X} = 20$

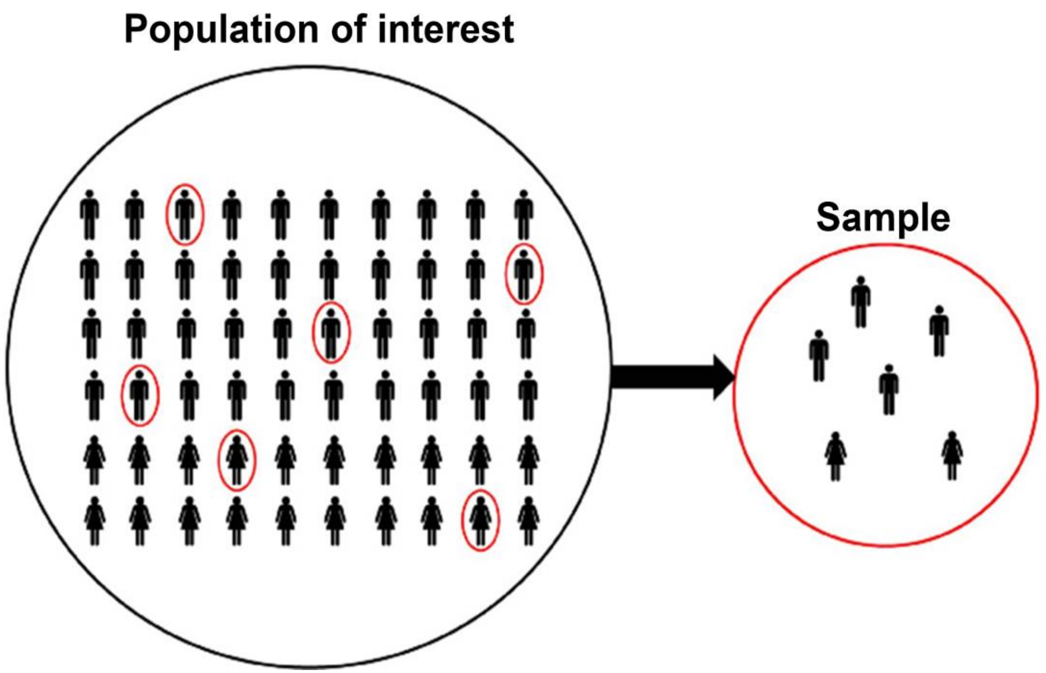
# Population VS. Sampling

# Population Vs. Sampling

- **The primary task of inferential statistics** is making an inference about something by using only an incomplete sample of data.

- **Population**: is a collection of all possible individuals, objects, or measurements of interest (To understand the whole collection of data).

- **Sample:** is a portion or part of the population of interest (To make inferences about the whole based on the subset).

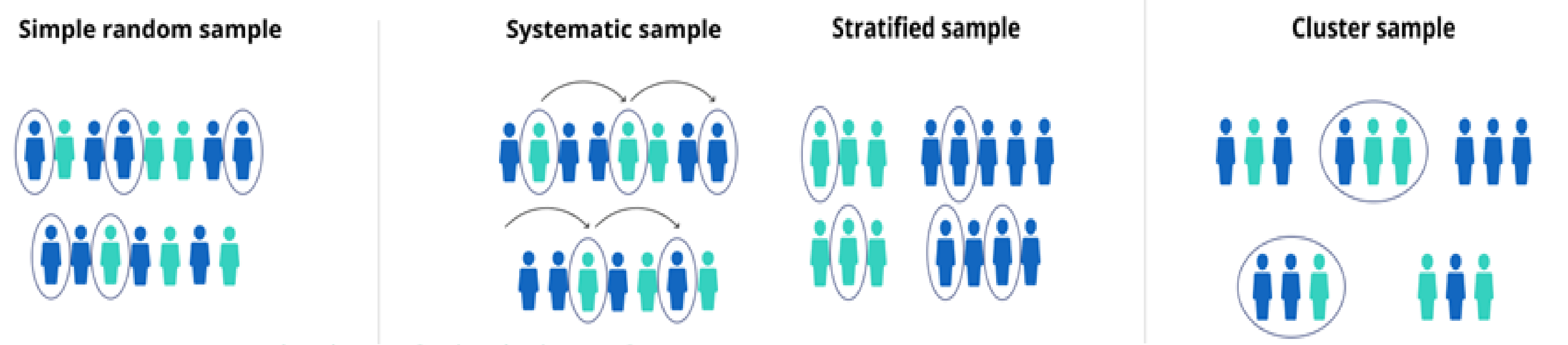


# Sampling

**Definition:** The process of selecting a subset of individuals from a population to estimate characteristics of the whole population.

**Types of Sampling:**

- **Random Sampling:** Every member has an equal chance of being selected.
- **Stratified Sampling:** Population divided into subgroups, and samples are drawn from each.
- **Systematic Sampling:** Selecting every nth member from a list.
- **Convenience Sampling:** Selecting individuals who are easiest to reach, which may introduce bias.



# When Should Samples be used?

- **When studying a large population** where it is impractical or impossible to collect data from every individual.

- **When resources** such as time, cost, and manpower are limited, making it more feasible to collect data from a subset of the population.

- **When conducting research or experiments** where it is important to minimize potential biases in data collection.

**For samples:**

$$\text{variance} = s^2 = \frac{\Sigma (x - \bar{x})^2}{n - 1}$$

$$\text{standard deviation} = s = \sqrt{s^2}$$

**Calculating Formula**

$$s^2 = \frac{\Sigma x^2 - \frac{(\Sigma x)^2}{n}}{n - 1}$$

$$S^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$$

**For populations:**

$$\text{variance} = \sigma^2 = \frac{\Sigma (x - \bar{x})^2}{n}$$

$$\text{standard deviation} = \sigma = \sqrt{\sigma^2}$$

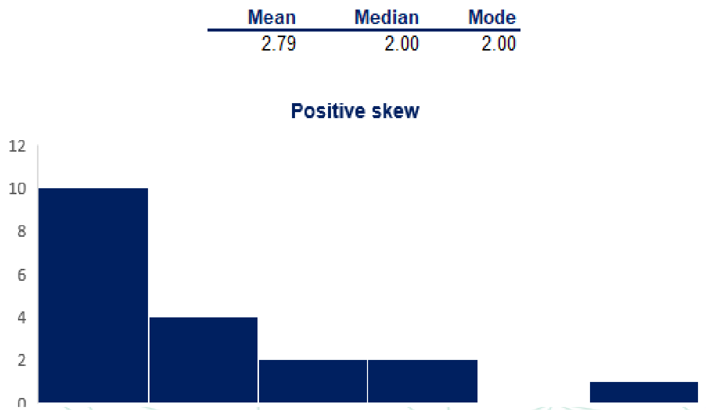
**Calculating Formula**

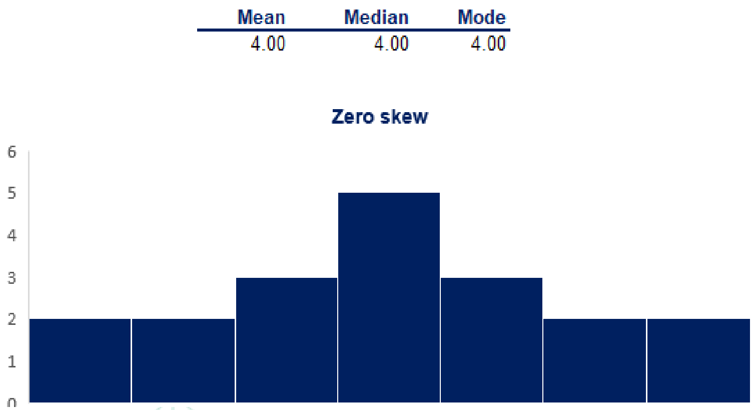
$$\sigma^2 = \frac{\Sigma x^2 - \frac{(\Sigma x)^2}{n}}{n}$$

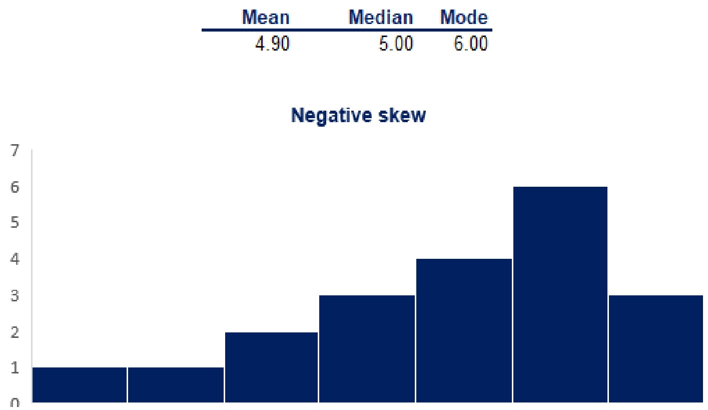
$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

# What is Skewness of Data?

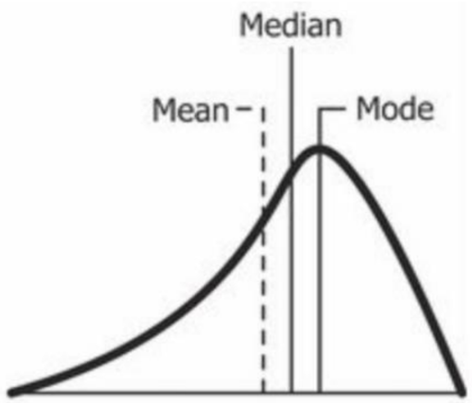
- Skewness is the measure of how much the probability distribution of a random variable deviates from the normal distribution.
- Skewness is positive if the tail of the distribution extends more to the right, and negative if the tail extends more to the left.

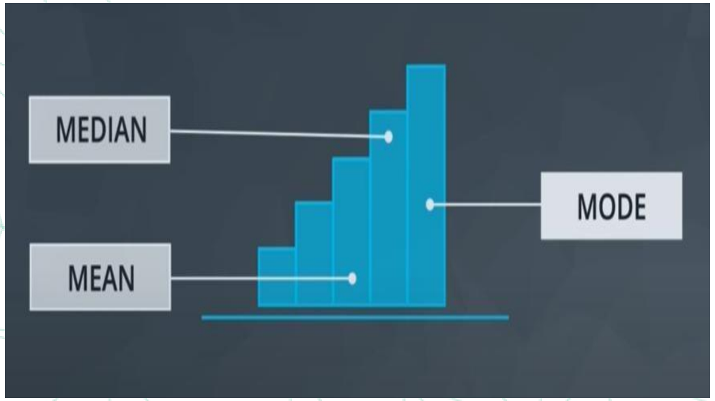


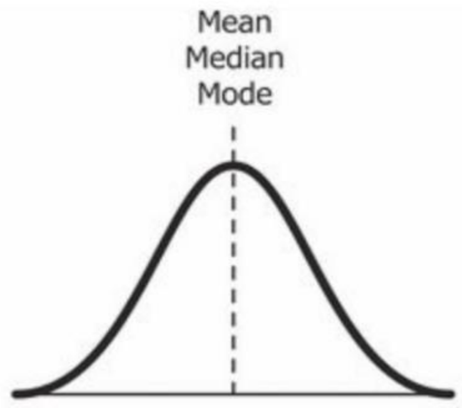
# Mean, Median and Mode

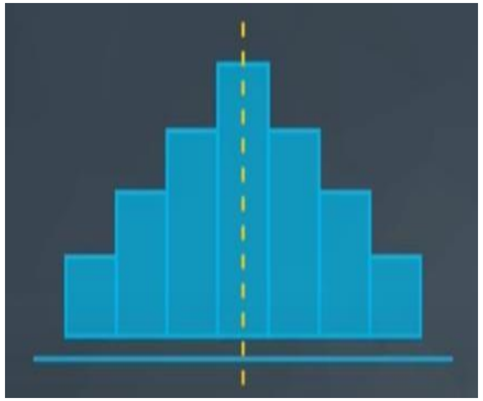



The value of skewness for a left skewed distribution is less than zero.

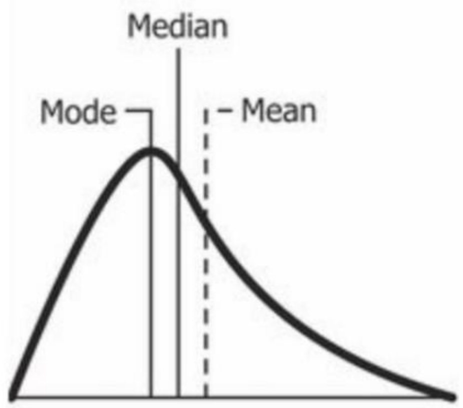
**mean < median < mode**

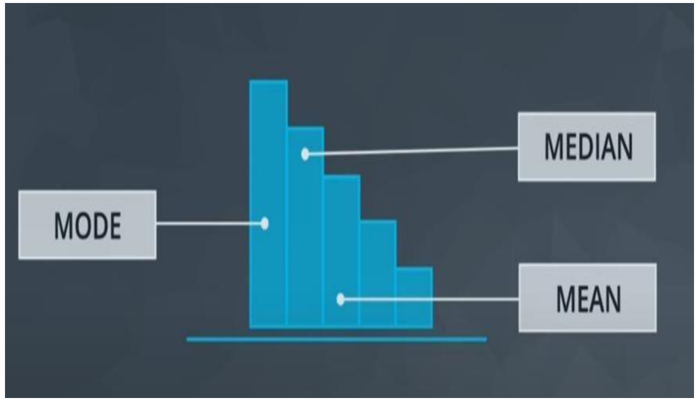


The value of skewness for a normal distribution is zero.

**mode = median = mean**




The value of skewness for a right skewed distribution is less than zero

**mode < median < mean**

# Correlation

- Used to find the relationship between two variables which is important in real because we can predict value of one variable.

Example from our daily life:

- The more time you spend running on a treadmill, the more calories you will burn.
- The more money you save, the more financially secure you feel.
- The more cigarettes you smoked, the higher stress level you have

Use Scatter plot in visualization of correlation

# Correlation Coefficient

- The correlation coefficient (r)/pearson r indicates a measure of the strength of a relationship between two variables.
  Possible correlations range from +1 to –1.

  **Direction of correlation :**

  - A correlation of –1 indicates a perfect negative correlation, meaning that as one variable goes up, the other goes down.
  - A correlation of +1 indicates a perfect positive correlation , meaning that as one variable goes up, the other goes up together.
  - A correlation of zero indicates that there is no relationship between the variables.

- Describe linear relations only

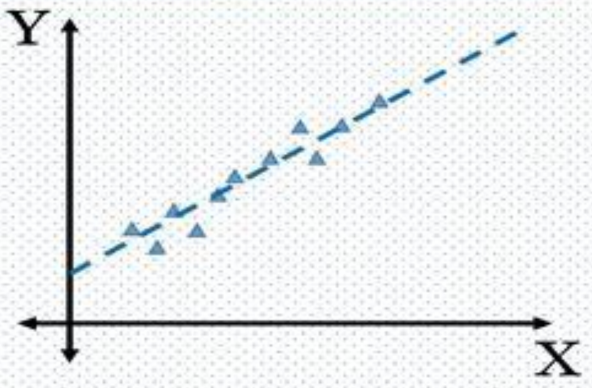
# Correlation Coefficient

## Statistically Correlated

- Strength of the correlation – Coefficient of Correlation
- Direction of correlation – Sign of the Coefficient

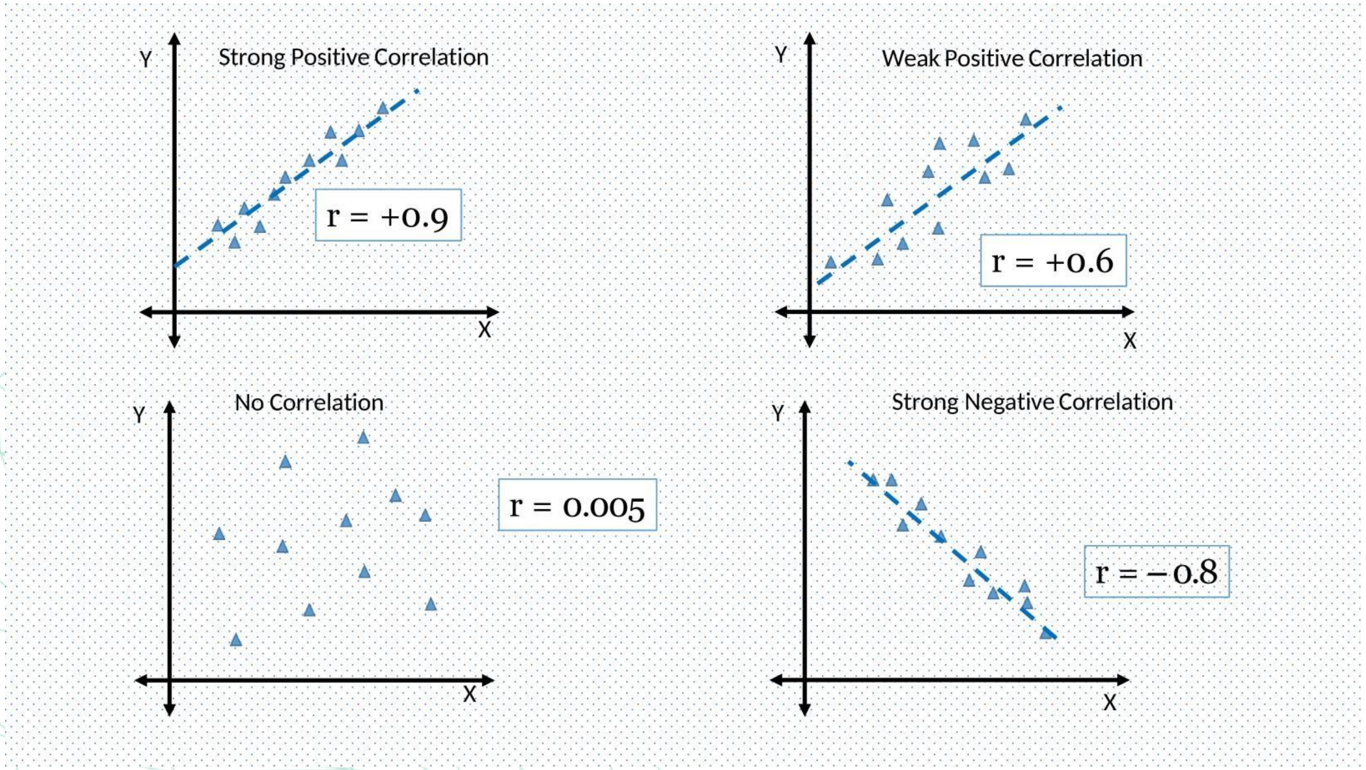
Pearson Correlation Coefficient

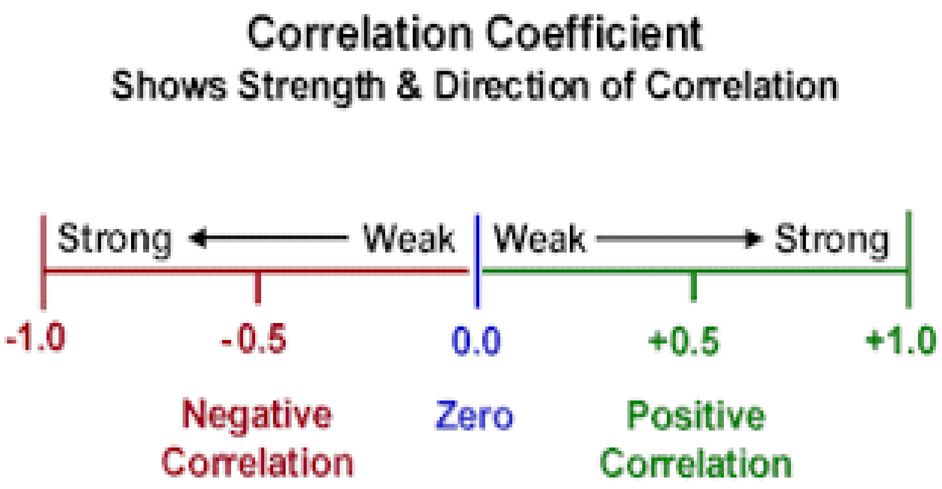
$$r = \frac{\sum (x - \bar{x}) * (y - \bar{y})}{(N - 1) * \sigma_x * \sigma_y}$$




- **X**: Values of the **x-variable** in a sample
- **x̄**: Mean of the values of the **x-variable**
- **y**: Values of the **y-variable** in a sample
- **ȳ**: Mean of the values of the **y-variable**
- **N**: Number of records
- **σ**: Standard deviation

# Correlation Coefficient




Scatter plot in visualization of correlation , The horizontal axis represents one variable, and the vertical axis represents the other.

**Regression line :** is the line separate the 2 classes As points surrounded regression line the strength of correlation increase

The closer the correlation is to 0, the weaker it is, while the closer it is to +/-1, the stronger it is.

# Correlation Coefficient

| | Height X | Weight Y | $X - \bar{X}$ | $Y - \bar{Y}$ | $(X - \bar{X}) * (Y - \bar{Y})$ |
|---|---|---|---|---|---|
| | 160 | 130 | -15.625 | -40.625 | 634.7656 |
| | 170 | 150 | -5.625 | -20.625 | 116.0156 |
| | 165 | 145 | -10.625 | -25.625 | 272.2656 |
| | 180 | 190 | 4.375 | 19.375 | 84.76563 |
| | 175 | 175 | -0.625 | 4.375 | -2.73438 |
| | 190 | 210 | 14.375 | 39.375 | 566.0156 |
| | 185 | 180 | 9.375 | 9.375 | 87.89063 |
| | 180 | 185 | 4.375 | 14.375 | 62.89063 |
| **Mean** | **175.625** | **170.625** | | | **1821.875** |
| **Std Dev** | **10.155** | **25.651** | | | |

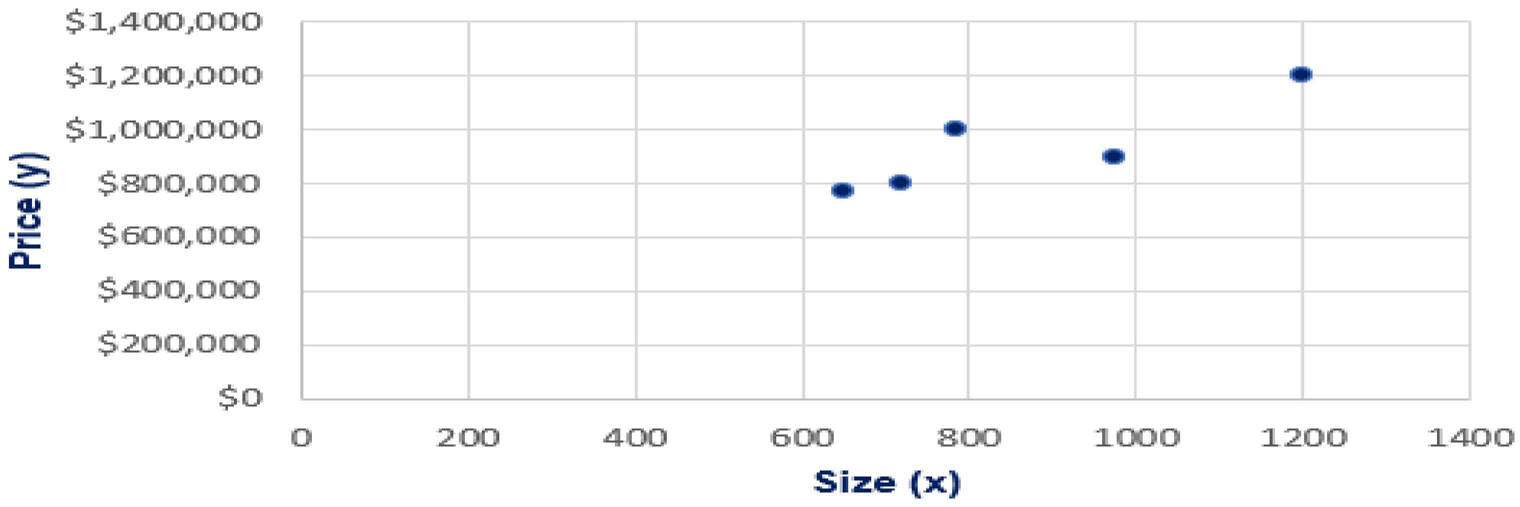
$$r = \frac{\sum (x - \bar{x}) * (y - \bar{y})}{(N - 1) * \sigma_x * \sigma_y}$$

$$r = \frac{1821.875}{(8-1) * 10.155 * 25.651}$$

$$r = 0.96$$

# Example: Housing Data

| | Size (ft.) | Price ($) | | $(x-\bar{x})*(y-\bar{y})$ |
|---|---|---|---|---|
| | 650 | 772,000 | | 34,776,000 |
| | 785 | 998,000 | | - 5,265,000 |
| | 1200 | 1,200,000 | | 89,178,000 |
| | 720 | 800,000 | | 19,418,000 |
| | 975 | 895,000 | | - 4,142,000 |
| **Mean** | 866 | 933,000 | **Sum** | 133,965,000 |
| | | | **Sample size** | 5 |
| | | | **Cov. Sample** | 33,491,250 |
| | | | **Corr Coefficient** | 0.87 |



# Task

Calculate Mean ,Median ,Mode ,Variance ,Std for these students in Math and Science Subjects Grades

| Student | Math | Science |
|---|---|---|
| A | 85 | 78 |
| B | 90 | 88 |
| C | 78 | 84 |
| D | 92 | 91 |
| E | 88 | 76 |

# Task Solution

Calculate Mean ,Median ,Mode ,Variance ,Std for these students in Math and Science Subjects Grades

## Math

Mean = (85+90+78+92+88)/5 = 86.6

Median = [78, 85, 88, 90, 92]

Mode = No mode (all grades are unique)

Variance = 23.84 Std = 4.88

Because

85 – 86.6 = -1.6 (-1.6)^2 = 2.56

90 - 86.6 = 3.4 (3.4 )^2 = 11.56 variance = (2.56 + 11.56 + 73.96 + 29.16 + 1.96)/5 = 23.84

78 - 86.6 = -8.6 (-8.6)^2 = 73.96 Std = √ 23.84 = 4.88

92 - 86.6 = 5.4 (5.4)^2 = 29.16

88 - 86.6 = 1.96 (1.96)^2 = 1.96

# Task Solution

Calculate Mean ,Median ,Mode ,Variance ,Std for these students in Math and Science Subjects Grades

## Science

Mean = (78+88+84+91+76)/5 = 83.4

Median = [76,78,84,88,91]

Mode = No mode (all grades are unique)

Variance = 32.64 Std = 5.71

Because

78 – 83.4 = -5.4 (-5.4)^2 = 29.16

88 - 83.4 = 4.6 (4.6 )^2 = 21.16 variance = (29.16 + 21.16 + 0.36 + 57.76 + 54.76)/5 = 32.64

84 - 83.4 = 0.6 (0.6)^2 = 0.36 Std = √ 32.64 = 5.71

91 - 83.4 = 7.6 (7.6)^2 = 57.76

76 - 83.4 = -7.4 (-7.4)^2 = 54.76

# Inferential Statistics

Data Distribution

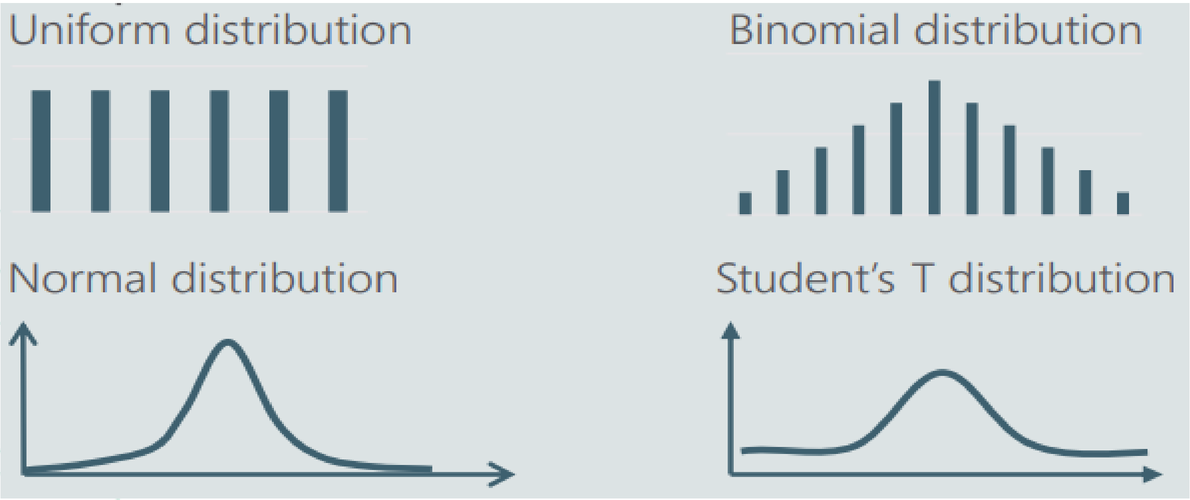
# Data Distribution

- A distribution is a function that shows the possible values for a variable and how often they occur
- In probability theory and statistics, a probability distribution is a mathematical function that provides the probabilities of the occurrence of different possible outcomes in an experiment
- It is a common mistake to believe that the distribution is the graph. In fact, the distribution is the ‘rule’ that determines how values are positioned in relation to each other. Very often, we use a graph to visualize the data. Since different distributions have a particular graphical representation, statisticians like to plot them.

# Probability Distribution

Common distributions include:

- **Normal Distribution**: Symmetrical, bell-shaped distribution characterized by its mean and standard deviation.
- **Binomial Distribution**: Represents the number of successes in a fixed number of trials with a constant probability of success.
- **Poisson Distribution**: Models the number of events occurring in a fixed interval of time or space.



# Normal Distribution

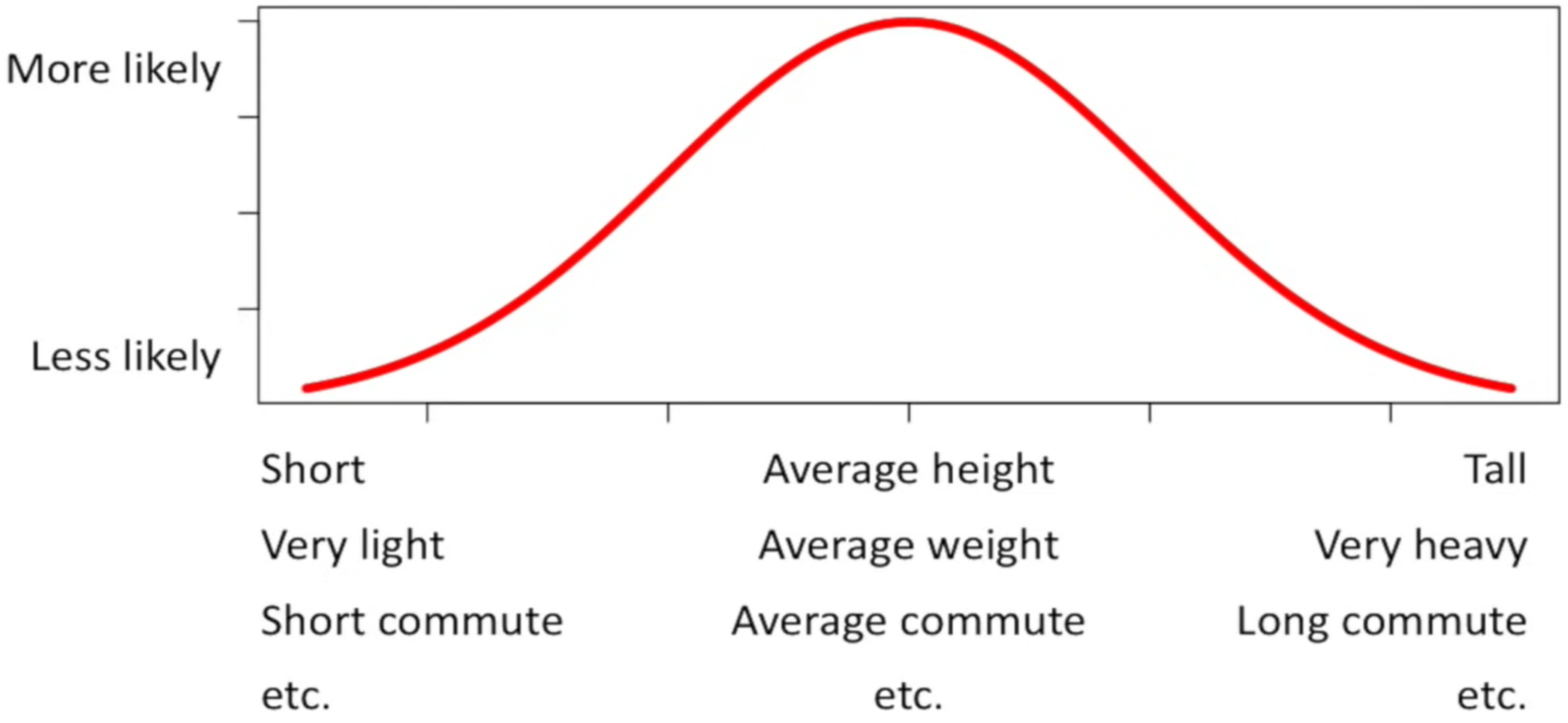
- **Normal distribution**, also known as the Gaussian distribution, is a probability distribution that is symmetric about the mean, showing that data near the mean are more frequent in occurrence than data far from the mean (Bell shape).

- **The Importance of Normal Distribution**
  1. It approximates a wide variety of random variables
  2. Distributions of sample means with large enough sample sizes could be approximated as normal
  3. All computable statistics are well-structured
  4. It is widely used in regression analysis
  5. Good track record

- **Applications:**
  - **Biology:** Most biological measures are normally distributed, such as: height, arm and leg length, nails, blood pressure, thickness of tree barks, etc.
  - **IQ tests**, **Stock Market Information**

# Normal Distribution

A lot of things are “normally distributed”.



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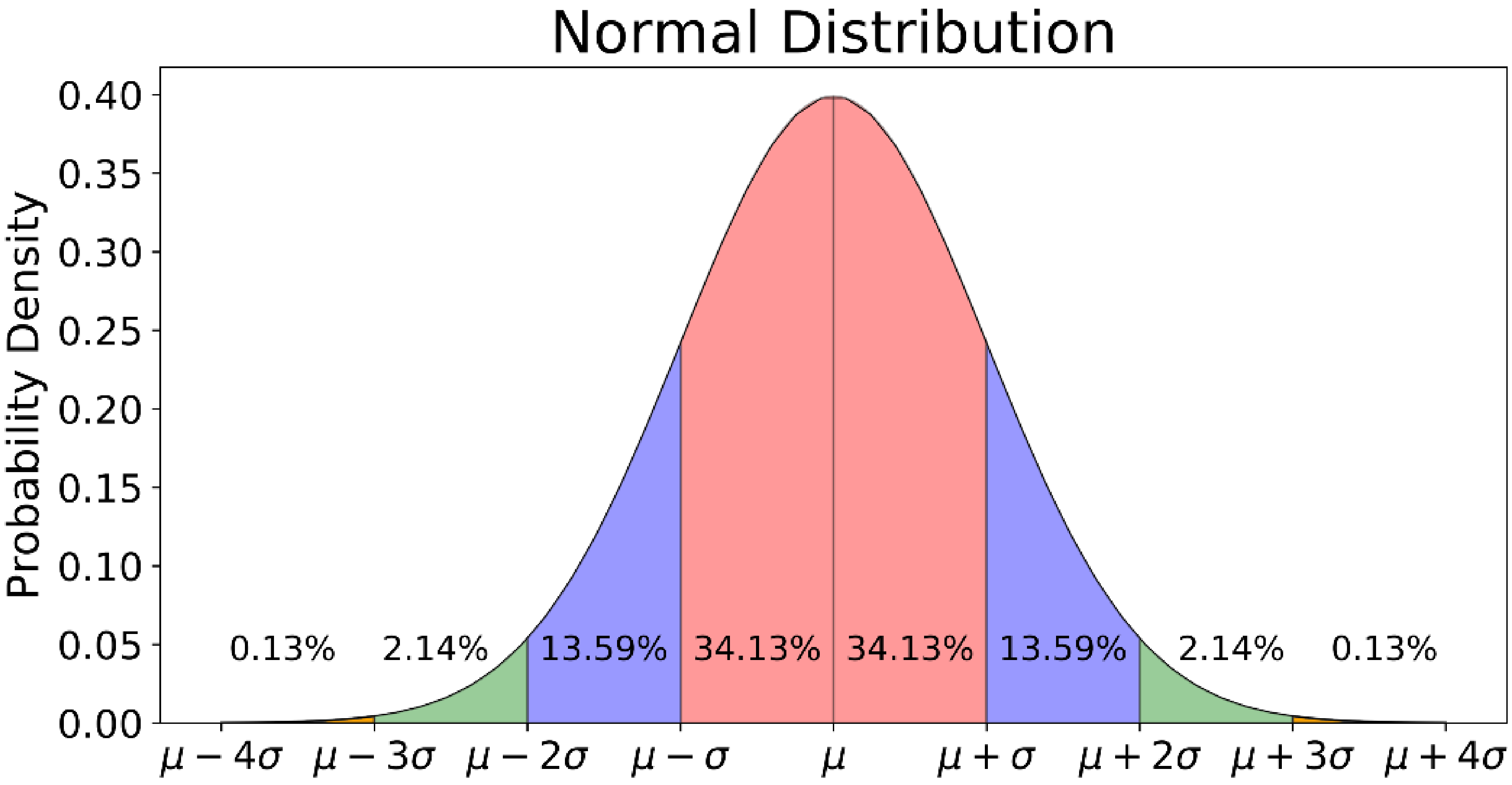
رواد مصر الرقمية

# How is Data Distributed in Normal Distribution?



# Hypothesis Testing

- A statistical method to make inferences or draw conclusions about a population based on sample data.

- Elements of a hypothesis test:

  - **Population:** The entire group being studied.
  - **Sample:** A subset of the population used to make inferences.
  - **Null Hypothesis ($H0$ ):** The default assumption (no effect or no difference).
  - **Alternative Hypothesis ($H1$ ):** The assumption that there is an effect or difference.
  - **Significance Level ($\alpha$):** The threshold for rejecting $H0$ (commonly 0.05).

# Hypothesis Testing

| Test Result – True State | $H_0$ True | $H_0$ False |
|---|---|---|
| $H_0$ True | Correct Decision | Type I Error |
| $H_0$ False | Type II Error | Correct Decision |

$$\alpha = P(Type\ I\ \ Error) \quad \beta = P(Type\ II\ \ Error)$$

**Conclusion.** If $H_0$ is rejected, we conclude that $H_A$ is true. If $H_0$ is not rejected, we conclude that $H_0$ may be true.

# Steps in Hypothesis Testing

**1.State the Hypotheses**:

1. **Null Hypothesis ($H0$ )**: This is the hypothesis that there is no effect or no difference. It represents the status quo or a statement of no change.
2. **Alternative Hypothesis ($H1$)**: This hypothesis represents what you aim to prove, indicating the presence of an effect or a difference.

**2.Choose the Significance Level (α$\alpha$)**:

- The significance level is the probability of rejecting the null hypothesis when it is true. Common choices are 0.05, 0.01, or 0.10.

**3.Select the Appropriate Test**:

1. Choose a statistical test based on the data type and the hypotheses. Common tests include:
   1. t-test (for comparing means)
   2. Chi-square test (for categorical data)
   3. ANOVA (for comparing means across multiple groups)
   4. Z-test (for large sample sizes)

**4.Collect Data**:

- Gather the sample data that will be used for the hypothesis test. Ensure the data is collected in a way that is unbiased and representative of the population.

# Steps in Hypothesis Testing

5. **Calculate the Test Statistic**:
   - Using the chosen statistical test, compute the test statistic (e.g., t-value, z-value) based on the sample data.
6. **Determine the p-value**:
   - The p-value indicates the probability of observing the test results under the null hypothesis. It helps to determine the strength of the evidence against the null hypothesis.
7. **Make a Decision**:
   1. Compare the p-value to the significance level ($\alpha$):
      1. If p≤α$p \leq \alpha$: Reject the null hypothesis ($H0$).
      2. If p>α$p > \alpha$: Fail to reject the null hypothesis ($H0$).
8. **Draw Conclusions**:
   - Interpret the results in the context of the research question. Discuss whether the evidence supports the alternative hypothesis and what it implies for the population.
9. **Report the Results**:
   - Clearly present the findings, including the hypotheses, test statistic, p-value, and any relevant confidence intervals. Provide context and implications of the results.

- Testing hypothesis for the mean μ :
  - When the value of sample size (n):

population is normal or not normal
( n ≥ 30 )

σ is known

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

σ is not known

$$Z = \frac{\bar{x} - \mu_0}{S / \sqrt{n}}$$

population is normal
(n< 30)

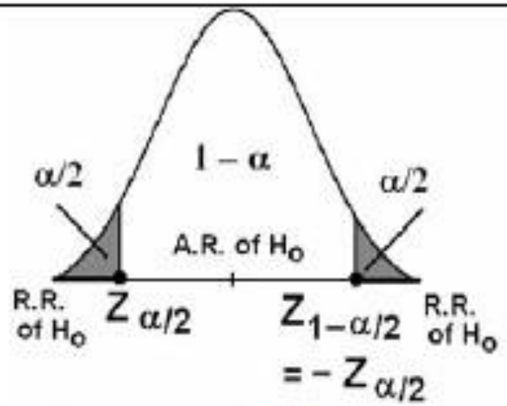
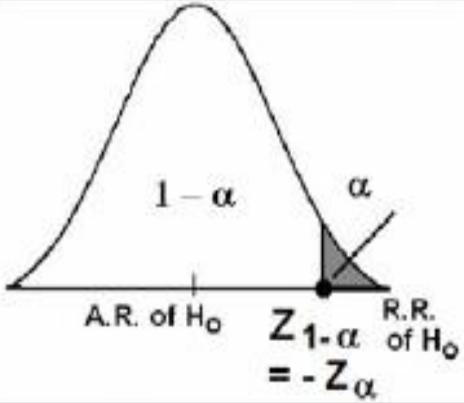
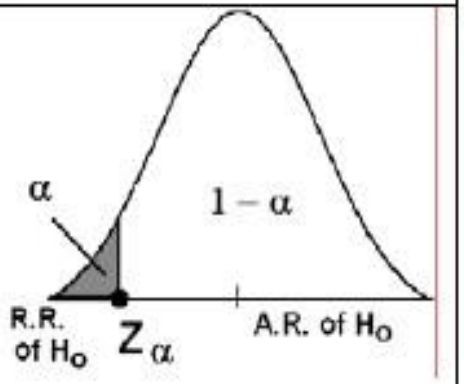
σ is known

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

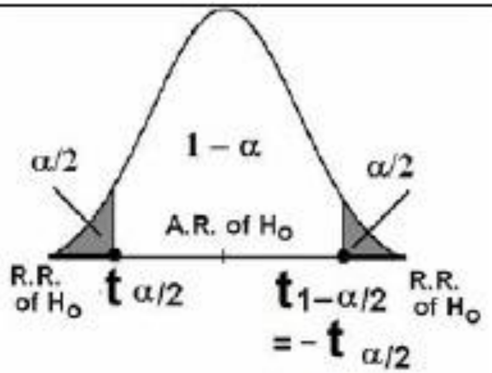
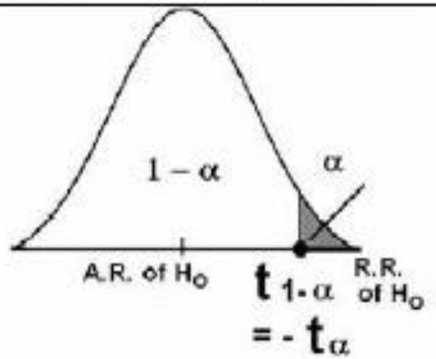
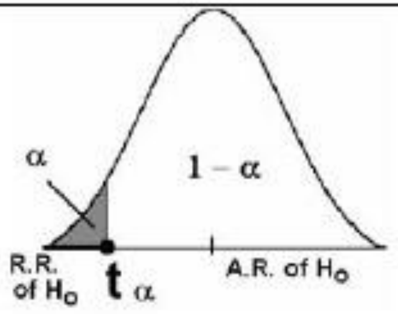
σ is not known

$$T = \frac{\bar{x} - \mu_0}{S / \sqrt{n}}$$

## Test Procedures:

| | | | |
|---|---|---|---|
| Hypotheses | $H_o: \mu = \mu_o$<br>$H_A: \mu \neq \mu_o$ | $H_o: \mu \leq \mu_o$<br>$H_A: \mu > \mu_o$ | $H_o: \mu \geq \mu_o$<br>$H_A: \mu < \mu_o$ |
| Test Statistic (T.S.) | Calculate the value of: $Z = \dfrac{\overline{X} - \mu_o}{\sigma / \sqrt{n}} \sim N(0,1)$ | | |
| R.R. & A.R. of $H_o$ |   |   |   |
| Critical value (s) | $Z_{\alpha/2}$ and $-Z_{\alpha/2}$ | $Z_{1-\alpha} = -Z_{\alpha}$ | $Z_{\alpha}$ |
| Decision: | We reject $H_o$ (and accept $H_A$) at the significance level $\alpha$ if: | | |
| | $Z < Z_{\alpha/2}$ or<br>$Z > Z_{1-\alpha/2} = -Z_{\alpha/2}$<br>Two-Sided Test | $Z > Z_{1-\alpha} = -Z_{\alpha}$<br>One-Sided Test | $Z < Z_{\alpha}$<br>One-Sided Test |

## Test Procedures:

| | | | |
|---|---|---|---|
| Hypotheses | $H_o: \mu = \mu_o$<br>$H_A \; \mu \neq \mu_o$ | $H_o: \mu \leq \mu_o$<br>$H_A: \mu > \mu_o$ | $H_o: \mu \geq \mu_o$<br>$H_A: \mu < \mu_o$ |
| Test Statistic (T.S.) | Calculate the value of: $t = \dfrac{\overline{X} - \mu_o}{S / \sqrt{n}} \sim t(n-1)$<br>$(df = \nu = n-1)$ | | |
| R.R. & A.R. of $H_o$ |  |  |  |
| Critical value (s) | $t_{\alpha/2}$ and $-t_{\alpha/2}$ | $t_{1-\alpha} = -t_{\alpha}$ | $t_{\alpha}$ |
| Decision: | We reject $H_o$ (and accept $H_A$) at the significance level $\alpha$ if: | | |
| | $t < t_{\alpha/2}$ or<br>$t > t_{1-\alpha/2} = -t_{\alpha/2}$<br>Two-Sided Test | $t > t_{1-\alpha} = -t_{\alpha}$<br>One-Sided Test | $t < t_{\alpha}$<br>One-Sided Test |

# Find the Critical t-value

## *t* Table

| cum. prob | $t_{.50}$ | $t_{.75}$ | $t_{.80}$ | $t_{.85}$ | $t_{.90}$ | $t_{.95}$ | $t_{.975}$ | $t_{.99}$ | $t_{.995}$ | $t_{.999}$ | $t_{.9995}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **one-tail** | **0.50** | **0.25** | **0.20** | **0.15** | **0.10** | **0.05** | **0.025** | **0.01** | **0.005** | **0.001** | **0.0005** |
| **two-tails** | **1.00** | **0.50** | **0.40** | **0.30** | **0.20** | **0.10** | **0.05** | **0.02** | **0.01** | **0.002** | **0.001** |
| **df** | | | | | | | | | | | |
| 1 | 0.000 | 1.000 | 1.376 | 1.963 | 3.078 | 6.314 | 12.71 | 31.82 | 63.66 | 318.31 | 636.62 |
| 2 | 0.000 | 0.816 | 1.061 | 1.386 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 | 22.327 | 31.599 |
| 3 | 0.000 | 0.765 | 0.978 | 1.250 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 | 10.215 | 12.924 |
| 4 | 0.000 | 0.741 | 0.941 | 1.190 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 | 7.173 | 8.610 |
| 5 | 0.000 | 0.727 | 0.920 | 1.156 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 | 5.893 | 6.869 |
| 6 | 0.000 | 0.718 | 0.906 | 1.134 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 | 5.208 | 5.959 |
| 7 | 0.000 | 0.711 | 0.896 | 1.119 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 | 4.785 | 5.408 |
| 8 | 0.000 | 0.706 | 0.889 | 1.108 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 | 4.501 | 5.041 |
| 9 | 0.000 | 0.703 | 0.883 | 1.100 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 | 4.297 | 4.781 |
| 10 | 0.000 | 0.700 | 0.879 | 1.093 | 1.372 | 1.812 | 2.228 | 2.764 | 3.169 | 4.144 | 4.587 |
| 11 | 0.000 | 0.697 | 0.876 | 1.088 | 1.363 | 1.796 | 2.201 | 2.718 | 3.106 | 4.025 | 4.437 |
| 12 | 0.000 | 0.695 | 0.873 | 1.083 | 1.356 | 1.782 | 2.179 | 2.681 | 3.055 | 3.930 | 4.318 |
| 13 | 0.000 | 0.694 | 0.870 | 1.079 | 1.350 | 1.771 | 2.160 | 2.650 | 3.012 | 3.852 | 4.221 |
| 14 | 0.000 | 0.692 | 0.868 | 1.076 | 1.345 | 1.761 | 2.145 | 2.624 | 2.977 | 3.787 | 4.140 |
| 15 | 0.000 | 0.691 | 0.866 | 1.074 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 | 3.733 | 4.073 |
| 16 | 0.000 | 0.690 | 0.865 | 1.071 | 1.337 | 1.746 | 2.120 | 2.583 | 2.921 | 3.686 | 4.015 |
| 17 | 0.000 | 0.689 | 0.863 | 1.069 | 1.333 | 1.740 | 2.110 | 2.567 | 2.898 | 3.646 | 3.965 |
| 18 | 0.000 | 0.688 | 0.862 | 1.067 | 1.330 | 1.734 | 2.101 | 2.552 | 2.878 | 3.610 | 3.922 |
| 19 | 0.000 | 0.688 | 0.861 | 1.066 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 | 3.579 | 3.883 |
| 20 | 0.000 | 0.687 | 0.860 | 1.064 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 | 3.552 | 3.850 |
| 21 | 0.000 | 0.686 | 0.859 | 1.063 | 1.323 | 1.721 | 2.080 | 2.518 | 2.831 | 3.527 | 3.819 |
| 22 | 0.000 | 0.686 | 0.858 | 1.061 | 1.321 | 1.717 | 2.074 | 2.508 | 2.819 | 3.505 | 3.792 |
| 23 | 0.000 | 0.685 | 0.858 | 1.060 | 1.319 | 1.714 | 2.069 | 2.500 | 2.807 | 3.485 | 3.768 |
| 24 | 0.000 | 0.685 | 0.857 | 1.059 | 1.318 | 1.711 | 2.064 | 2.492 | 2.797 | 3.467 | 3.745 |
| 25 | 0.000 | 0.684 | 0.856 | 1.058 | 1.316 | 1.708 | 2.060 | 2.485 | 2.787 | 3.450 | 3.725 |
| 26 | 0.000 | 0.684 | 0.856 | 1.058 | 1.315 | 1.706 | 2.056 | 2.479 | 2.779 | 3.435 | 3.707 |
| 27 | 0.000 | 0.684 | 0.855 | 1.057 | 1.314 | 1.703 | 2.052 | 2.473 | 2.771 | 3.421 | 3.690 |
| 28 | 0.000 | 0.683 | 0.855 | 1.056 | 1.313 | 1.701 | 2.048 | 2.467 | 2.763 | 3.408 | 3.674 |
| 29 | 0.000 | 0.683 | 0.854 | 1.055 | 1.311 | 1.699 | 2.045 | 2.462 | 2.756 | 3.396 | 3.659 |
| 30 | 0.000 | 0.683 | 0.854 | 1.055 | 1.310 | 1.697 | 2.042 | 2.457 | 2.750 | 3.385 | 3.646 |
| 40 | 0.000 | 0.681 | 0.851 | 1.050 | 1.303 | 1.684 | 2.021 | 2.423 | 2.704 | 3.307 | 3.551 |
| 60 | 0.000 | 0.679 | 0.848 | 1.045 | 1.296 | 1.671 | 2.000 | 2.390 | 2.660 | 3.232 | 3.460 |
| 80 | 0.000 | 0.678 | 0.846 | 1.043 | 1.292 | 1.664 | 1.990 | 2.374 | 2.639 | 3.195 | 3.416 |
| 100 | 0.000 | 0.677 | 0.845 | 1.042 | 1.290 | 1.660 | 1.984 | 2.364 | 2.626 | 3.174 | 3.390 |
| 1000 | 0.000 | 0.675 | 0.842 | 1.037 | 1.282 | 1.646 | 1.962 | 2.330 | 2.581 | 3.098 | 3.300 |
| ***z*** | 0.000 | 0.674 | 0.842 | 1.036 | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 | 3.090 | 3.291 |
| | 0% | 50% | 60% | 70% | 80% | 90% | 95% | 98% | 99% | 99.8% | 99.9% |
| | **Confidence Level** | | | | | | | | | | |

# Example 1 - Efficacy Test for New drug

- Drug company has new drug, wishes to compare it with current standard treatment
- Federal regulators tell company that they must demonstrate that new drug is better than current treatment to receive approval
- Firm runs clinical trial where some patients receive new drug, and others receive standard treatment
- Numeric response of therapeutic effect is obtained (higher scores are better).
- Parameter of interest: $\mu_{\text{New}} - \mu_{\text{Std}}$

رواد مصر الرقمية

وزارة الاتصالات
وتكنولوجيا المعلومات

# Example 1 - Efficacy Test for New drug

- **Null hypothesis -** New drug is no better than standard trt

$$H_0 : \mu_{New} - \mu_{Std} \leq 0 \qquad \left(\mu_{New} - \mu_{Std} = 0\right)$$

- **Alternative hypothesis -** New drug is better than standard trt

$$H_A : \mu_{New} - \mu_{Std} > 0$$

- **Experimental (Sample) data:**

| | |
|---|---|
| $\bar{y}_{New}$ | $\bar{y}_{Std}$ |
| $s_{New}$ | $s_{Std}$ |
| $n_{New}$ | $n_{Std}$ |

# Example 1 - Efficacy Test for New drug

- **Type I error -** Concluding that the new drug is better than the standard ($H_A$) when in fact it is no better ($H_0$). Ineffective drug is deemed better.
    - Traditionally $\alpha$ = P(Type I error) = 0.05

- **Type II error -** Failing to conclude that the new drug is better ($H_A$) when in fact it is. Effective drug is deemed to be no better.
    - Traditionally a clinically important difference ($\Delta$) is assigned and sample sizes chosen so that:

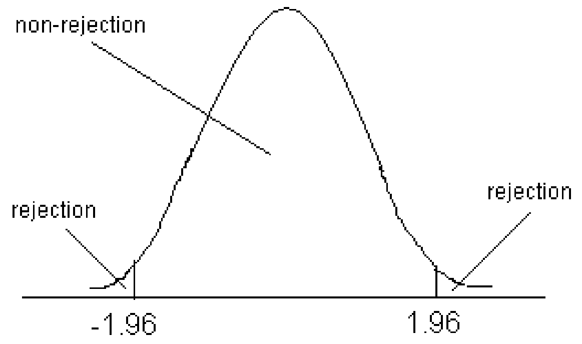
      $\beta$ = P(Type II error | $\mu_1$-$\mu_2$ = $\Delta$) $\leq$ .20

# Example 2 - Mean Age of a Certain Population

- Researchers are interested in the mean age of a certain population.
- A random sample of 10 individuals drawn from the population of interest has a mean of 27.
- Assuming that the population is approximately normally distributed with variance 20,can we conclude that the mean is different from 30 years? ($\alpha$=0.05) .
- If the p - value is 0.0340 how can we use it in making a decision?

# Solution

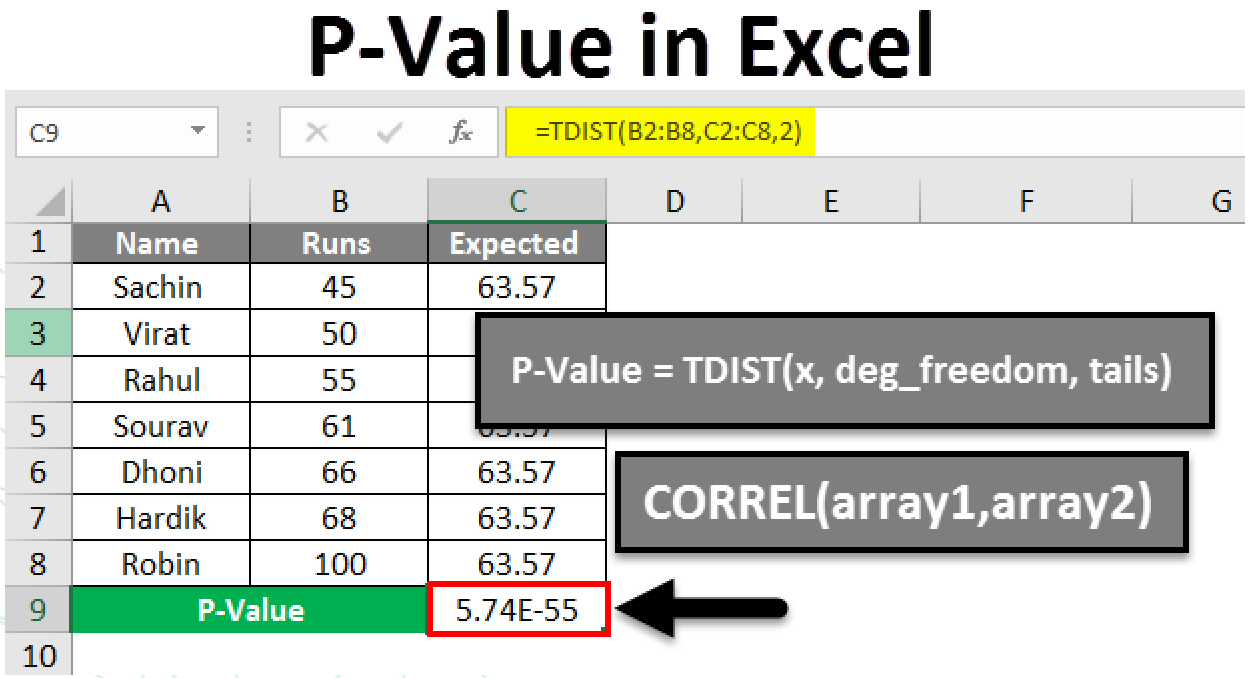
1. **Data:** variable is age, n=10, x =27 ,$\sigma^2$=20,α=0.05
2. **Assumptions:**
   The population is approximately normally distributed with variance 20
3. **Hypotheses:**
- ➢ $H_0$ : μ=30
- ➢ $H_A$ : μ ≠30
4. **Test Statistic:** Z= -2.12
5. **Decision Rule**
- ➢ The alternative hypothesis is $H_A$: μ ≠ 30
- ➢ Hence we reject $H_0$ if $Z > Z_{1-0.025} = Z_{0.975}$ or $Z < -Z_{1-0.025} = -Z_{0.975}$
- ➢ $Z_{0.975}$=1.96(from table D)




6. **Decision:**
- ➢ We reject $H_0$ ,since -2.12 is in the rejection region .
- ➢ We can conclude that μ is not equal to 30
- ➢ Using the p value ,we note that p-value =0.0340 < 0.05,therefore we reject $H_0$

# T.DIST Function

We can use Excel's **T.DIST Function** to calculate the p-value by simply adding the test statistic value and degree of freedom



# Regression

# Regression

Regression is a statistical method used to understand the relationship between variables. The primary purpose of regression analysis is to predict a dependent variable (also known as the outcome, response, or target) based on one or more independent variables (also known as predictors, features, or explanatory variables).

**Types of Regression:**

- ➢ **Simple Linear Regression:** One independent variable, straight-line relationship.
- ➢ **Multiple Linear Regression:** Multiple independent variables predicting one dependent variable.
- ➢ **Logistic Regression:** Used for predicting binary outcomes.

# Types of Regression

# Linear Regression

- **Linear Regression**: The simplest form of regression, where the relationship between the dependent and independent variable(s) is modeled as a straight line. It can be represented by the equation

$$Y = \beta_0 + \beta_1 x + \epsilon$$

where:

- $Y$ is the dependent variable.
- $\beta_0$ is the intercept.
- $\beta_1$ is the slope (coefficient of the independent variable $x$).
- $\epsilon$ is the error term.

**Predicting House Prices**:

**Scenario**: Predicting the price of a house based on a single feature, such as square footage.
**Example**: Using historical data on house prices and their square footage to build a model that estimates the price of a new house based on its size.

# Multiple Linear Regression

- **Multiple Linear Regression**: An extension of linear regression that uses more than one independent variable. The model can be written as

$$\mathrm{Y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_{\mathrm{n}} x_{\mathrm{n}} + \epsilon$$

where:

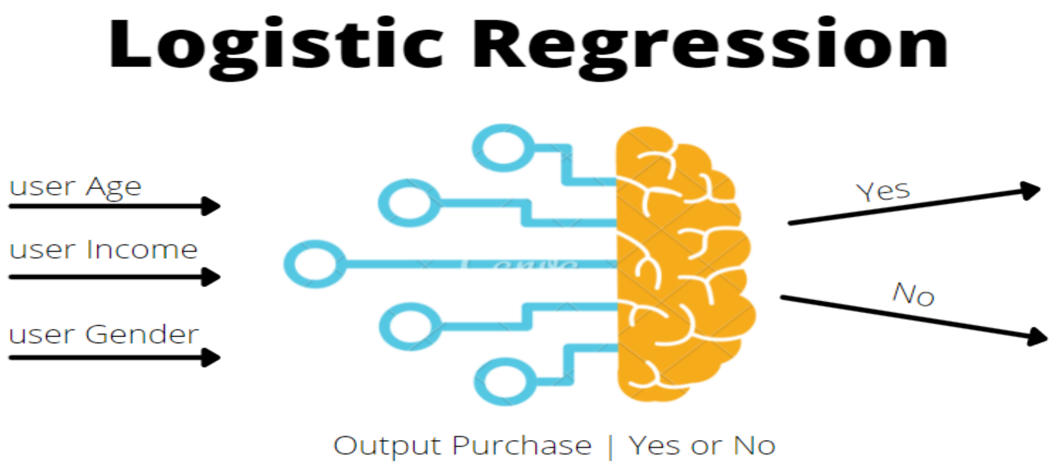
- $Y$ is the dependent variable.
- $\beta_0$ is the intercept.
- $\beta_1$ is the slope (coefficient of the independent variable $x$).
- $\epsilon$ is the error term.

**Predicting Salary:**

**Scenario**: Estimating an employee’s salary based on multiple factors, such as years of experience, education level, and job role.
**Example**: Using a dataset that includes years of experience, education level (Bachelor’s, Master’s, PhD), and job roles (Developer, Manager, Analyst) to predict salaries.

# Logistic Regression

Used for binary classification problems, where the dependent variable is categorical (e.g., success/failure, yes/no). It uses the logistic function to model the probability of the default class.




**Spam Detection:**

**Scenario:** Classifying emails as spam or not spam.

**Example:** Using logistic regression to build a spam filter that identifies emails as spam based on features like the presence of certain keywords, sender information, and email structure.

# Questions?